

FUNDAMENTALS OF CHEMISTRY (LAB MANUAL)



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Triton College

Fundamentals of Chemistry (Lab manual)

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CHAPTER OVERVIEW

0: LAB 0 - LAB SAFETY ORIENTATION

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0.1: Lab Safety Videos

Safety Orientation Videos

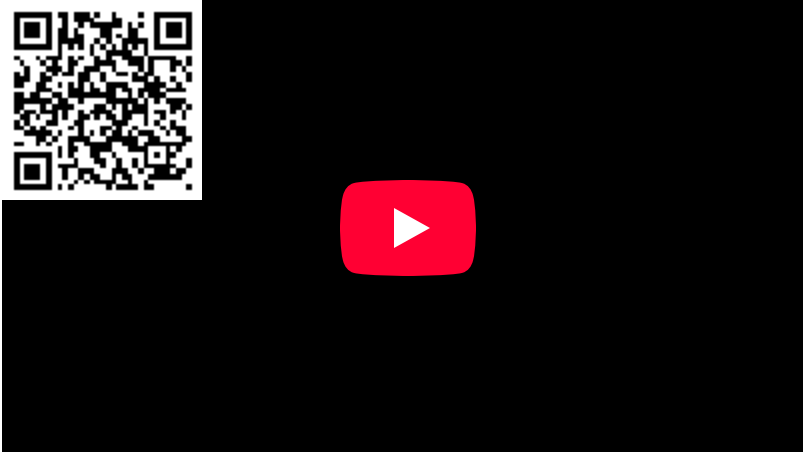
Students will find below the eight Lab safety orientation videos. The first six are part of the College Lab Safety Series by the American Chemical Society. Students should watch the first seven videos before signing the safety contract and attempting the safety quiz. The eighth video is an older version of the College Lab Safety video by the American Chemical Society, which should be watched, time permitting, as it provides detailed dos and don'ts in a college lab setting.

1) ACS (American Chemical Society) College Lab Safety Series





2) Lab Safety and Techniques- Crash course safety



3) [Safety Video by American Chemical Society](#) (1991)



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0.2: Lab Safety Contract

Triton College Chemistry Division

Student Safety Contract

Purpose

At Triton College, safety is our number one priority. A list of rules has been developed and provided to you in this student safety contract to ensure a safe science laboratory experience. These rules must always be followed. Please be advised that you will be asked to leave the laboratory if you violate these rules.

No exceptions!!

This contract is available in both PDF and .doc formats. Download, sign, and date both contract pages before you participate in the laboratory sessions. Your signature indicates that you have read this contract and agree to abide by the rules. You will keep a copy for your records and upload or submit one to your instructor via your Blackboard or Learning Management System (LMS) as directed.

I, _____, dated _____, agree that I will comply with the following safety rules. I will:

- 1) Always wear safety goggles in the laboratory.
- 2) Always wear appropriate clothing in the laboratory. This includes long pants, shirts with sleeves, and closed shoes (i.e., no sandals). Also, I understand that long hair must be tied back, and loose or baggy clothing or jewelry must be secured.

***Note: Students violating rules one and/or two will be asked to leave the lab immediately. No second chances!! NO EXCEPTIONS!!**

- 3) Read laboratory instructions beforehand and ask the instructor about any part of the lab I am unsure about.
- 4) Use proper equipment for laboratory procedures.
- 5) Never perform unauthorized experiments or work alone in the laboratory.
- 6) Never remove chemicals or other materials/equipment from the laboratory area.
- 7) Read the labels on reagent bottles carefully, remove only small amounts of reagent with the proper tools, and never return unused chemicals to the bottle.
- 8) Dispose of broken glass and waste chemicals in the appropriate waste container.
- 9) Immediately inform the instructor of a chemical spill or accident (even minor ones) in the laboratory.
- 10) Never taste or smell any chemicals.
- 11) Know the location of eyewash fountains, fire extinguishers, fire blankets, safety showers, first aid kits, and exits.
- 12) Do not bring food or drink into the lab; refrain from eating or drinking in the laboratory. This includes chewing gum and eating candy.
- 13) Keep aisles clear. Books, purses, backpacks, and other personal belongings will be kept outside or under the lab benches.
- 14) I will responsibly conduct myself at all times in the laboratory. Horseplay, practical jokes, and pranks are dangerous and prohibited.
- 15) Clean my work area before leaving the lab.
- 16) Wash my hands before I leave the lab.

Agreement

I, _____, have read and agree to follow all the safety rules outlined in this contract. I realize I must obey these rules to ensure my safety and that of my fellow students. I will fully cooperate with my instructor and fellow students to maintain a safe lab environment. I will also closely follow all oral and written instructions provided by the instructor, the science

department, and the college. I am aware that any violation of this safety contract, resulting in unsafe conditions in the laboratory or misbehavior on my part, may lead to my removal from the laboratory, a failing grade, and/or dismissal from the course.

Student Signature

Date

Course section and number

Questions

1) Are you color blind? Yes/No

2) Do you have any allergies? Yes/No

If yes, please list specific allergies.

3) Do you wear contact lenses? Yes/No

4) If you are pregnant, breastfeeding, or trying to conceive, avoid working with heavy metal compounds and organic solvents. Notify your instructor and request a list of chemicals used in this course. Use this list to consult with your physician before proceeding with the course. Under your physician's supervision, you may need to purchase a respirator, which a doctor must fit and adjust to your needs.

5) Please inform your instructor of any other medical conditions or situations you feel we should be aware of to ensure your safety and the safety of others while working in the laboratory. Please be aware that anything written in this contract will remain confidential.

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0.3: LAB SAFETY QUIZ

Multiple-Choice Questions

1. What should you do if a chemical splashes into your eyes?
 - A) Rub your eyes vigorously
 - B) Rinse your eyes with water for at least 15 minutes
 - C) Close your eyes and wait for the pain to subside
 - D) Apply a cold compress
2. Which of the following is NOT appropriate lab attire?
 - A) Closed-toe shoes
 - B) Long pants
 - C) Loose, flowing clothing
 - D) Safety goggles
3. What is the first thing you should do if you spill a chemical on your skin?
 - A) Wipe it off with a cloth
 - B) Rinse the affected area with water immediately
 - C) Apply a bandage
 - D) Ignore it if it doesn't hurt
4. Where should you dispose of broken glassware?
 - A) Regular trash bin
 - B) Recycling bin
 - C) Designated glass disposal container
 - D) Sink
5. Which of the following statements about lab safety is TRUE?
 - A) Eating and drinking are allowed in the lab
 - B) You should always work alone in the lab
 - C) Always read labels and safety data sheets before using chemicals
 - D) It's okay to leave a Bunsen burner unattended

True/False Questions

6. Always wear safety goggles when handling chemicals.
 - True
 - False
 7. It's safe to taste chemicals in the lab to identify them.
 - True
 - False
 8. You should be familiar with the location of safety equipment, including fire extinguishers, eyewash stations, and safety showers.
 - True
 - False
 9. When working in the lab, it is recommended to tie long hair back.
 - True
 - False
 10. Using your phone in the lab is acceptable as long as you're careful.
 - True
 - False
-

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0.4: ADDITIONAL RESOURCES

OSHA® QUICK CARD™

Hazard Communication Standard Pictogram

As of June 1, 2015, the Hazard Communication Standard (HCS) will require pictograms on labels to alert users of the chemical hazards to which they may be exposed. Each pictogram consists of a symbol on a white background framed within a red border and represents a distinct hazard(s). The pictogram on the label is determined by the chemical hazard classification.

HCS Pictograms and Hazards

Health Hazard  • Carcinogen • Mutagenicity • Reproductive Toxicity • Respiratory Sensitizer • Target Organ Toxicity • Aspiration Toxicity	Flame  • Flammables • Pyrophorics • Self-Heating • Gases Flammable Gas • Self-Reactives • Organic Peroxides	Exclamation Mark  • Irritant (skin and eye) • Skin Sensitizer • Acute Toxicity (Harmful) • Narcotic Effects • Respiratory Tract Irritant • Hazardous to Ozone Layer (Non-Mandatory)
Gas Cylinder  • Gases Under Pressure	Corrosion  • Skin Corrosion/ Burns • Eye Damage • Corrosive to Metals	Exploding Bomb  • Explosives • Self-Reactives • Organic Peroxides
Flame Over Circle  • Oxidizers	Environment (Non-Mandatory)  • Aquatic Toxicity	Skull and Crossbones  • Acute Toxicity (fatal or toxic)

For more information:
OSHA® Occupational Safety and Health Administration
 U.S. Department of Labor
www.osha.gov (800) 321-OSHA (6742)

OSHA 3091 (v.10)

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1: LAB 1 - INTRODUCTION TO CHEMISTRY AND SCIENTIFIC METHOD

PURPOSE

The purpose of this experiment is to:

- Introduction to Chemistry
- Demonstrate skill in handling glassware and transferring chemicals.
- Analyze the observations and record the data.

INTRODUCTION

Chemistry is the branch of science that deals with the study of matter, its properties, composition, structure, and the changes it undergoes. It is a fundamental science crucial in understanding the world around us. Chemistry has numerous applications in various fields, including medicine, materials science, agriculture, and environmental science.

The study of science is not just about knowing but also about asking questions, making predictions, testing ideas, and drawing meaningful conclusions. In Chemistry, we do not just mix substances and observe changes. Chemists employ the **Scientific method**, a structured approach that involves observation, hypothesis, experimentation, and conclusion.

The scientific method is the systematic approach chemists and scientists use to investigate natural and chemical changes, propose explanations based on evidence, and test those explanations through experiments. **The scientific method** typically involves the following steps:

Observation: Scientists observe and gather information about natural or chemical phenomena or ask questions based on their observations.

Hypothesis: A hypothesis is a proposed explanation for the observed phenomenon. It is a testable statement that can be supported or rejected through experimentation and observation.

Experimentation: Scientists design experiments to test the hypothesis. Experiments involve controlled conditions to isolate variables and gather data that support or refute the hypothesis.

Data Collection and Analysis: Data is collected and analyzed during the experiment. Data collection and analysis involve statistical analysis and evaluation to determine if the results support the hypothesis.

Conclusion: Based on the data obtained, scientists conclude whether the hypothesis is supported or rejected. If supported, the hypothesis may become a theory or a scientific law. If not, it may be revised, refined, or discarded.

In this lab experiment, you will practice and explore using the scientific method by performing the experiments.

1) Effect of Heat

When heated, some compounds change, releasing a colorless gas. We will investigate how heat affects the behavior of substances and explore the resulting changes that occur.

✓ Example 1.1

Effect of heat on light green ferrous sulfate crystals ($\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$).

Solution

Upon gentle heating, a spatula of light green ferrous sulfate crystals ($\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$) in a test tube, the light green crystals change to white, and a colorless liquid is seen. (If further heated, the solid changes to brown.)

Table 1.1: Effect of Heat

Observation	Hypothesis
Colorless liquid initially seen upon heating the light green crystals	Heating Light green crystals produces water

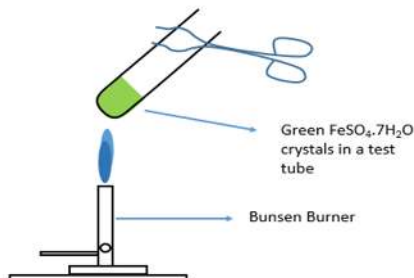
Light Green crystals changed to colorless solid

The light green crystals change into a white solid by losing water.

Colorless liquid initially seen upon heating the light green crystals

Heating Light green crystals produces water

Light Green crystals changed to colorless :



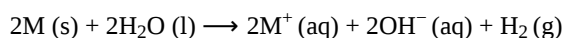
tals change into a white solid by losing

Figure 1.1: Heating Green Crystals of $(\text{FeSO}_4 \cdot 7\text{H}_2\text{O})$. (Copyright; Saadia Khan)

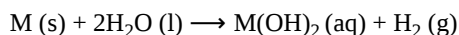
2) Effect of Water

Some elements are more reactive with water than others. A common characteristic of most Alkali Metals (Group 1) and most of the Alkaline Earth Metals (Group 2) is their ability to displace $\text{H}_2(\text{g})$ from water. Alkali metals are also known to react **violently and explosively with water**.

The general reaction of an alkali metal (M) with $\text{H}_2\text{O}(\text{l})$ is given in the following equation:



Most of the Alkaline Earth Metals (except Beryllium (Be)) also produce hydroxides when reacted with water. The general reaction of calcium, strontium, and barium with water is represented below, where M represents calcium, strontium, or barium:



✓ Example 1.2

Interaction of Magnesium Metal (Mg) with water (H_2O)

Solution

A strip of magnesium ribbon, approximately 1 to 2 inches long, is added to a hot water test tube. Magnesium (Mg) reacts with hot water to release clear gas ($\text{H}_2(\text{g})$). (Reaction of Magnesium with steam produces Magnesium oxide and hydrogen gas.)

Table 1.2: Effect of Water

Observation	Hypothesis
Bubbles are seen adhering to the magnesium ribbon, and these bubbles emerge from the solution as a colorless gas.	Colorless gas is produced during the reaction.
The solution turns pink when a few drops of Phenolphthalein indicator are added. If litmus paper is used, the solution indicates a basic pH.	As the reaction proceeds, a base or basic chemical is produced.

3) Reactivity

Comparing the reactivities of two metals. Some elements are more reactive than others.

A **chemical reactivity series** helps determine which elements are more reactive or less, which you will learn later in the course when studying chemical reactions.

✓ Example 1.3

Identify if the metals (Mg, Al) are reactive or unreactive in 1.0 M HCl.

Solution

To observe which metals are more reactive or unreactive, add 10.0 – 15.0 mL of 1.0 M hydrochloric acid to two test tubes. Cut two inches of Mg ribbon and Al foil and add them to the test tubes.

Table 1.3: Reactivity

Observation	Hypothesis
When Mg ribbon is added to a test tube containing HCl, it shows signs of reaction instantly, releasing a clear gas as vigorous bubbles.	Mg metal is a more reactive metal with acids.
All metals start reacting with the acid after nearly all the magnesium has reacted.	Al metal also reacts with acid, but it is slower than Mg metal.

4) Solubility

Some substances are soluble in water, while others are not. Different compounds have different solubilities. Compounds such as Ammonium chloride and silver chloride are dissolved in deionized water, and their solubilities are observed.

✓ Example 1.4

Solubilities of ammonium chloride (NH_4Cl) and silver chloride (AgCl), in deionized water.

Solution

Add approximately 10.0–15.0 mL of deionized water to each of two test tubes. Using a spatula, add a small amount of ammonium chloride (NH_4Cl) and silver chloride (AgCl), respectively, and observe their solubilities.

Table 1.4: Solubility

Observation	Hypothesis
Add a small amount of ammonium chloride, NH_4Cl , to the first test tube and dissolve it in distilled water.	Ammonium chloride is highly soluble in water.
In the second test tube containing distilled water, when a small amount of silver chloride is added and stirred with a glass rod, a white solid remains in the solution and settles to the bottom of the test tube.	Silver chloride is not soluble in water.

5) Temperature

A temperature change affects the solubility of compounds and influences chemical reactions.

✓ Example 1.5

Demonstrate effect of temperature on solubility of compound (Lead (II) chloride, PbCl_2).

Solution

Take two test tubes and add 10.0-15.0 mL of distilled water to the test tubes. A small amount of Lead (II) chloride, PbCl_2 , is added to both the test tubes, and their solubilities are observed. Place a 150.0 mL beaker containing approximately 100.0 mL of water on a hot plate to create a hot bath. Turn the hot plate off when the water boils, or use the hot water bath provided by the instructor. Take test tube two and place it in the hot water bath, then observe.

Table 1.5: Temperature

Observation

In test tube one, 10.0 – 15.0 mL of distilled water at room temperature is added to a small amount of lead (II) chloride, PbCl_2 , and stirred using a glass rod. Even after stirring it for a few minutes, PbCl_2 does not dissolve.

Test tube two contains 10.0 – 15.0 mL of distilled water at room temperature. A small amount of lead (II) chloride, PbCl_2 , is added and stirred using a glass rod. After stirring it for a few minutes, PbCl_2 does not dissolve. Place test tube two into the hot water bath, and lead chloride, PbCl_2 , dissolves.

Hypothesis

Lead (II) chloride, PbCl_2 , is not soluble in distilled water at room temperature.

Lead (II) chloride, PbCl_2 , is not soluble in room-temperature water but dissolves in hot distilled water. An increase in the water's temperature helps dissolve the PbCl_2 .

EQUIPMENT* AND CHEMICALS NEEDED

Table 1.6: Equipment and Chemicals

Equipment	Equipment	Chemicals	Chemicals
Safety goggles	Bunsen burner and rubber tubing	Powdered Sugar	1.0 M Hydrochloric acid
Deionized Water or Distilled Water	Test tubes, test tube holder, test tube rack	Sodium carbonate \ Na_2CO_3	Calcium carbonate \ CaCO_3
Stirring rod with Rubber policeman	Celsius Thermometer	(Copper (II) sulfate pentahydrate $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$)	Calcium (Ca) metal
Graduated cylinder (10.0 mL or 25.0 mL).	Beakers: 100.0 mL; 150.0 mL.	Copper (Cu) metal	Zinc (Zn) metal

* Images of equipment needed in this lab can be found in the appendix (the equipment may differ or be subject to changes; follow your instructor's directions).



SAFETY PRECAUTIONS

1. Always wear chemical splash goggles during the experiment.
2. Gloves are provided for those who wish to wear them.
3. Extreme caution while using a Bunsen burner, Hydrochloric acid, and hot water
4. Dispose of waste in designated containers.
5. Waste of Copper sulfate and any leftover metals should be disposed of in proper waste containers. (as indicated during the experimental procedure by your instructor).
6. Clean the lab benches and verify that all equipment is in your lab drawer before leaving.
7. Ensure all glassware is clean and all equipment is returned to its designated place.
8. Wash your hands as soon as you leave the lab.

This Laboratory exploration encompasses a series of experiments focusing on the effects of various factors, including heat, water, reactivity, solubility, and temperature, on elements and compounds. You will record your observations while experimenting and suggest a hypothesis based on them.

EXPERIMENTAL PROCEDURE

1) Effect of Heat

Using a spatula, add a small amount of pale blue crystals of copper sulfate to a test tube or an evaporating dish and heat them. Use the flame of a Bunsen burner for the test tube or an evaporating dish. Using a hot plate, use an evaporating dish instead of a test

tube. Observe the changes, record your observations on the data sheet, and then state your hypothesis.

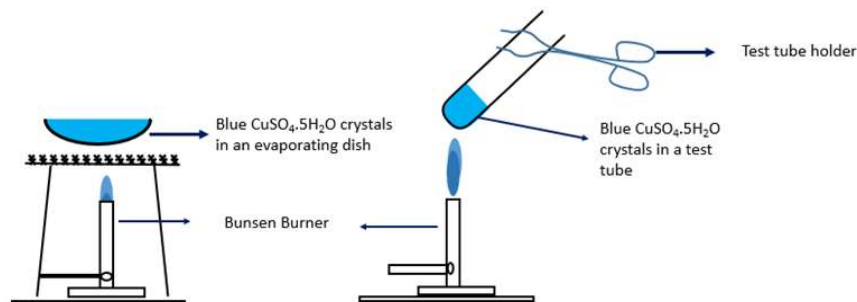


Figure 1.2: Heating Blue crystals of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$. (Copyright; Saadia Khan)

Let the test tube cool down and discard the solid into the inorganic waste container or as suggested by your instructor. If the solid becomes stuck in the test tube and cannot be removed with a spatula, place the test tube in a test tube rack. Add 10.0 mL of water and let the solid dissolve. Discard the bluish liquid into the inorganic waste container.

2) Effect of Water

Compare the effect of water on two different metals, Copper (Cu) and Calcium (Ca). Take two large test tubes available in your lab, add 10.0-15.0 mL of distilled water to each, and place them in a test tube rack. Label them one and two. Add a half-inch-long copper wire or a small piece of copper metal (Cu) in test tube one. On the datasheet, note down your observations and state your hypothesis.

Add a tiny piece of calcium (Ca) metal to the second test tube and observe the changes. On the datasheet, note your observations and state your hypothesis. Discard any metal into the designated metal waste container, as instructed by your instructor. (Pour the contents of the test tubes into 100. Use a 10 mL beaker and remove any unreacted metals using tweezers. Set them on a paper towel, and at the end of the experiment, dispose of them in the metal waste container.

3) Reactivity

We will compare the reactivity of two metals, copper (Cu) and zinc (Zn), in hydrochloric acid. Label two test tubes, '1' and '2', and add 10.0 mL of hydrochloric acid to each. Add a small piece of Zn metal to test tube one and Cu metal to test tube two, respectively. On the datasheet, note your observations and state your hypothesis. Discard any metal into the designated metal waste container, as instructed by your instructor. (Pour the contents of the test tubes into 100. Use a 10 mL beaker and remove any unreacted metals using tweezers. Set them on a paper towel, and at the end of the experiment, dispose of them in the metal waste container.

4) Solubility

Different compounds have different solubilities. Test the solubilities of compounds such as Sodium carbonate and Calcium carbonate. Label two test tubes, '1' and '2', and add 15.0 mL of deionized or distilled water to each. In test tube one, add a small amount of Sodium carbonate, Na_2CO_3 , using a spatula, and stir the mixture using a glass rod. In test tube 2, add a minimal amount of calcium carbonate (CaCO_3) and stir with a glass rod. On the datasheet, note your observations and state your hypothesis. Discard as suggested by your instructor.

5) Temperature:

Fill three test tubes with 10.0 – 15.0 mL of distilled water each. Place the first test tube in an ice bath, leave the second at room temperature, and place the third test tube in the hot water bath. Add one tablespoon of sugar to each of the three test tubes: into test tube one after the water has cooled to around 10-15 minutes, and into test tube three after the water has been heated but not boiled. On the datasheet, note your observations and state your hypothesis. Discard the contents as suggested by your instructor.

PRE-LAB QUESTIONS

Name: _____

1. What is the scientific method?

2. Why do chemists and scientists use the scientific method? Explain the significance of the scientific method in understanding chemical phenomena.

3. What safety precautions should be taken while performing this experiment?

4. For each of the **five experimental aspects** mentioned below, based on the examples you have studied, predict what may happen before you do the experiments in this lab. State a hypothesis.
 - **a) Effect of Heat:**

 - **b) Effect of Water:**

 - **c) Reactivity:**

 - **d) Solubility:**

 - **e) Temperature:**

DATA OBSERVATIONS AND CALCULATIONS

Name: _____ Lab Partner(s): _____

Data Table:

Table 1.7: Data Table

	Observations	Hypothesis
1. Effect of Heat:		
$\text{CuSO}_4 \cdot 5\text{H}_2\text{O} \xrightarrow{\Delta}$		
2. Effect of Water:		
$\text{Cu (s)} \xrightarrow{\text{H}_2\text{O}}$		
$\text{Ca (s)} \xrightarrow{\text{H}_2\text{O}}$		
3. Reactivity:		
$\text{Zn (s)} \xrightarrow{\text{HCl}}$		
$\text{Cu (s)} \xrightarrow{\text{HCl}}$		
4. Solubility:		
$\text{Na}_2\text{CO}_3 \text{ (s)} \xrightarrow{\text{H}_2\text{O}}$		
$\text{CaCO}_3 \text{ (s)} \xrightarrow{\text{H}_2\text{O}}$		
5. Temperature		
$\text{Sugar (s)} \xrightarrow{\text{H}_2\text{O (cold)}}$		
$\text{Sugar (s)} \xrightarrow[\text{(Room temperature)}]{\text{H}_2\text{O}}$		
$\text{Sugar (s)} \xrightarrow{\text{H}_2\text{O (hot)}}$		

POST-LAB QUESTIONS

1. Were the hypotheses you predicted for the experiments accurate? If not, which ones did you need to revise?

2. What are the steps of the scientific method? Did you carry out these steps in your experiment?

Please click [here](#) to access the Pre-Lab, Data Tables, and Post-Lab in Word or PDF format. Complete them and upload the lab report according to your instructor's instructions.

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2: LAB 2 - INSTRUMENTAL MEASUREMENTS

PURPOSE

The purpose of this experiment is to:

- Record measurements of length, mass, volume, and temperature.
- Measure the mass and volume of an unknown rectangular solid.
- Demonstrate proficiency in using the following instruments: metric rulers, balances, graduated cylinders, and thermometers.

INTRODUCTION

Chemistry, a fundamental branch of science, is an experimental science. It is a qualitative science based on measurements that depends on careful observations of experiments, yielding accurate and precise results through the use of good laboratory techniques and instruments or devices.

In the chemistry laboratory, you must precisely measure length, mass, volume, and temperature for various experiments. To reinforce your understanding of common laboratory concepts and techniques and to gain knowledge in data gathering, this laboratory will explore and master fundamental operations using standard instruments and devices, thereby enhancing your success.

Measurements consist of two parts: a numerical part and a unit (sometimes called a dimension).

Example:

10.32	cm	10.000	g
Numerical part	unit	Numerical part	unit

Please make it a habit to always include the unit with each number you use.

Measurements made are not always accurate. The size of the discrepancy depends on several factors, including the skill of the person performing the measurement and the instrument or device used for the measurement. The margin of error in each measurement refers to the uncertainty associated with that measurement. The instrument itself limits how close one can come to a perfect measurement.

The measurements need to be precise and accurate. **Exact** measurements are those that consistently give close results. **Accurate** results are those that are close to the actual value. Precise and accurate measurements require instruments that are properly calibrated and functioning correctly. Additionally, the person performing the measurement must be familiar with using the instrument and follow the procedure carefully. By doing repeated measurements, one can estimate the precision of the instrument and the operator.

Measurement of Length

Length measurements in chemistry laboratories are mostly done using rulers in metric units. The rulers (Figure 1) are marked with standard distances in centimeters at the top and inches at the bottom, and the intervals on the ruler are referred to as hash marks.

Example 2.1

Measure the length of the yellow rectangle using a ruler

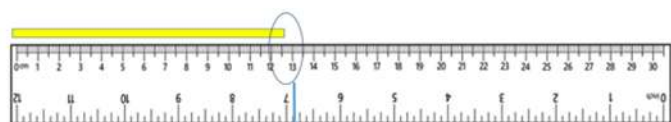
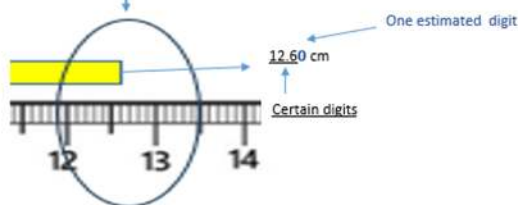


Figure 1: A ruler with Centimeters and inches markings used to measure yellow rectangle.



Solution

In measurements, all the figures recorded are significant, including certain digits plus the estimated digit. On Ruler, the yellow rectangle appears to be 12.6 cm long. You can estimate one digit between the scale markings and measure 12.60 cm.

Measurement of Mass

Another common and essential operation in the chemical laboratory is determining the mass of given objects or samples. For that purpose, a suitable balance must be used. In the lab, you will primarily use the milligram or analytical balances in the balance room. The analytical balances weigh the nearest 0.001 gram, and all the digits should be recorded for the exact mass.

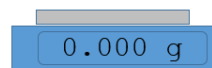


Figure 2. Milligram balance

Analytical balances are expensive and delicate, so care should be taken when using them. Always tare the balance, i.e., zero it before measuring the mass. Ensure the object you are measuring is always at room temperature, neither cold nor hot, to obtain an accurate mass measurement.

Many of the experimental procedures in this course require accurate quantitative values. In such instances, an analytical balance will be employed in weight determinations.

Measurement of Volume

Graduated cylinders, pipettes, or burettes are commonly used to measure volume. Volume in the laboratory is generally expressed in milliliters (mL) or cubic centimeters (cm³)

✓ Example 2.2

Measure the volume of water in the graduated cylinder in **Figure 3**.

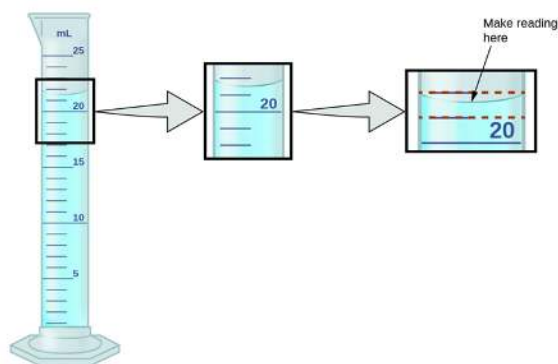


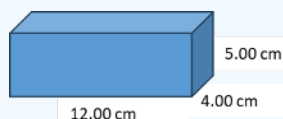
Figure 3: To measure the volume of liquid in this graduated cylinder.

Solution

The graduated cylinder with water above shows that the bottom of the meniscus lies between the 21 and 22 markings, indicating that the liquid volume is undoubtedly more than 21 mL but less than 22 mL. The meniscus appears closer to the 22-mL mark than the 21-mL mark, so a reasonable estimate of the liquid's volume would be 21.6 mL. In the number 21.6, the digits 2 and 1 are specific, but the 6 is an estimate.

✓ Example 2.3

Calculate the volume of a regular rectangular solid whose length, width, and height are measured to be 12.00 cm, 4.00 cm, and 5.00 cm, respectively.



Solution

Volume of a rectangular solid is $= L \times W \times h = 12.00 \text{ cm} \times 4.00 \text{ cm} \times 5.00 \text{ cm} = 240.00 \text{ cm}^3$

Measurement of Temperature

In chemistry laboratories, temperature measurement is performed using thermometers that contain colored liquid or are digital, typically in degrees Celsius. Other temperature scales that may be used are Fahrenheit and Kelvin.

Formulas for converting one temperature scale to another:

$$K = C + 273.15 \text{ } ^\circ\text{C} = \frac{(^{\circ}\text{F} - 32)}{1.8} \text{ } ^\circ\text{F} = (^{\circ}\text{C} \times 1.8) + 32.$$

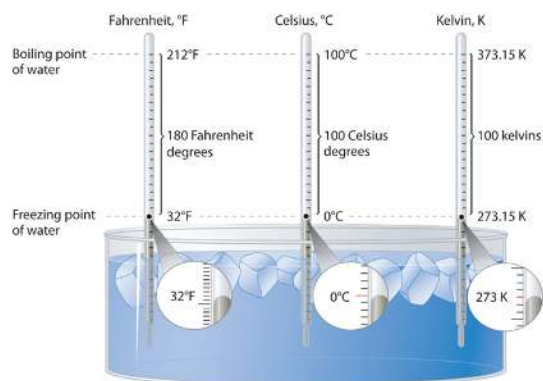


Figure 4: A Comparison of Fahrenheit (°F), Celsius (°C), and Kelvin (K) Temperature Scales showing the difference between the freezing point of water and the boiling point of water.

✓ Example 2.4

A student measures the temperature of boiling water to 99.5 °C. What would boiling water temperature be in degrees Fahrenheit (°F) and Kelvin?

Solution

$$^{\circ}\text{F} = (^{\circ}\text{C} \times 1.8) + 32 = (99.5 \text{ } ^{\circ}\text{C} \times 1.8) + 32 = 211.1 \text{ } ^{\circ}\text{F}$$

$$K = C + 273.15 = 99.5 \text{ } ^{\circ}\text{C} + 273.15 = 372.75 \text{ K}$$

EQUIPMENT* AND CHEMICALS NEEDED

Table 2.1: Equipment and Chemicals

Equipment	Equipment
Safety goggles	Large Test tube
Ruler - Centimeter and Inches	10.0 mL and 100.0 mL graduated cylinders.
Watch glass.	100.0 mL Beaker
Index card	Unknown Rectangular block
Milligram balance	Celsius Thermometer
Deionized water	

* Images of equipment needed in this lab can be found in the appendix (the equipment may differ or be subject to changes; follow your instructor's directions).

Safety Precautions

1. Always wear chemical splash goggles during the experiment.
2. Gloves are provided for those who wish to wear them.
3. Exercise extreme caution when using a hot plate or Bunsen burner to heat water.
4. Discard used water down the drain.
5. Ensure all glassware is clean and all equipment is returned to its designated place.
6. Clean the lab benches and verify all equipment is in your lab drawer before leaving.
7. Wash your hands as soon as you leave the lab.

EXPERIMENTAL PROCEDURE

A. Measurements of Length

1. Using the ruler, measure the length and width of the index card provided to your group in both inches and centimeters.
2. Measure the length of the 10.0 mL graduated cylinder in your lab drawer.
3. Measure the diameter of the watch glass in your lab drawer and compare it to that of other groups in your lab.

B. Measurements of Mass.

4. Using an analytical balance, measure the mass of the evaporating dish in your lab drawer.
5. Measure the mass of a watch glass in your lab drawer and compare it to that of other groups in your lab.
6. Measure the mass of a large test tube and compare it to that of other groups.

C. Measurements of Volume.

7. Take a 100.0 mL beaker, fill it with water to a volume of 50.0 mL, and pour it into a 100.0 mL graduated cylinder. Record the exact volume.
8. Fill the large test tube with water and record the exact volume by pouring it into the graduated cylinder.

D. Measurement of Mass and Volume of Unknown Solid.

9. Note the unknown rectangular solid color and the unknown number or letter.
10. Measure and record the exact mass of the solid using an analytical balance.
11. Using a ruler, measure and record its length, width, and height in cm.
12. Calculate the volume of the solid

E. Measurements of Temperature

14. Obtain a degree Celsius ($^{\circ}\text{C}$) thermometer.
15. Measure the room temperature and record the result.
16. Measure and record the temperature of cold tap water.
17. Measure and record the temperature of hot tap water (to hot/warm water provided by the instructor).

PRE-LAB QUESTIONS

Name: _____

1. In a chemistry laboratory, metric units are used for various measurements. Give the most commonly used metric unit for the measurements listed below:
 - a) Mass:
 - b) Length:
 - c) Temperature:
 - d) Volume
2. What are the **SI units** for the following measurements?
 - a) Mass:
 - b) Length:
 - c) Temperature:
 - d) Volume:
3. Normal human body temperature is 98.6 degrees Fahrenheit ($^{\circ}\text{F}$). **Convert** it to corresponding degrees Celsius ($^{\circ}\text{C}$) and Kelvin temperatures. (Show your work.)

DATA AND OBSERVATIONS

Name: _____ Lab Partner(s): _____

A. Measurements of Length.

A. Table 2.2: Measurements of Length

		Centimeters	Inches
1a.	Length of an Index card		
1b.	Width of an Index card		
2.	Length of the 10 mL graduated cylinder		
3.	Diameter of the watch glass		

4. Compare your measurements with those of another group in the lab. How do they compare and explain?

B. Measurements of Mass.

B. Table 2.3: Measurements of Mass

		Grams
1.	Mass of evaporating dish	
2.	Mass of a watch glass	
3.	Mass of a large test tube	

4. Compare your measurements with those of another group in the lab. How do they compare and explain?

C. Measurements of Volume.

1. Exact volume of water transferred from the beaker into the 100 mL graduated cylinder:
2. Were the volumes in the beaker and the graduated cylinder identical or similar? If not, explain:
3. Exact volume of water transferred from large test tube into graduated cylinder:

D. Measurement of Mass and Volume of Unknown Solid.

Unknown Number or Letter:

Description of Solid:

1. Mass of the unknown solid:
2. Length of unknown solid:
3. Width of unknown solid:
4. Height of unknown solid:
5. Volume of the solid (Show your work):

E. Measurements of Temperature

1. Room temperature:
2. Temperature of cold tap water:
3. Temperature of hot tap water:

POST LAB QUESTIONS

1. Perform the following conversions, show your work:

- a) 2.5 cm = _____ inch
- b) 5.2 L = _____ mL
- c) 5.25 kg = _____ g
- d) 25.2 °C = _____ F
- e) 27.2 °C = _____ K
- f) -21.0 °F = _____ °C

2. What is the volume of a rectangular wooden block with a length, width, and height of 5.5 cm, 4.5 cm, and 3.5 cm, respectively? Show your work.

3. Convert the room temperature, cold, and hot water readings into degrees Fahrenheit ($^{\circ}\text{F}$) and Kelvin (K). Show your work.

Please click here to access the Pre-Lab, Data Tables, and Post-Lab in Word or PDF format. Complete them and upload according to your instructor's instructions.

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3: LAB 3 - EXPLORING DENSITY

PURPOSE

The purpose of this experiment is to:

- Determine the density of water.
- Determine the density of an unknown liquid and infer its identity from a table of listed densities.
- Calculate the densities of regularly shaped and irregularly shaped solids.

INTRODUCTION

Density is defined as the amount of material present in a given volume. Every substance has a unique density value at a certain temperature, regardless of the amount of material present. (This is known as an intensive property.) For example, 25.0 mL and 50.0 mL of water should have a 1.00 g/mL density at room temperature because the mass-to-volume ratio is the same. Due to this nature, density can be used to infer the identity of an unknown substance.

In part of this activity, you will learn to experimentally determine water density. You will repeat the procedure with an unknown liquid and be asked to infer the unknown liquid's identity using a table of density values. Procedures to calculate the density of regularly and irregularly shaped solids will also be explored, providing a comprehensive understanding of density.

✓ Example 3.1

5.0 mL of water was added to a graduated cylinder with a mass of 20.175 g. The combined mass of the graduated cylinder and water was 25.180 g. Using this information, determine the experimental density of water.

Solution

To find the mass of water, we subtract the mass of the empty graduated cylinder from the combined mass of the graduated cylinder and the water, as shown below:

$$\text{Mass of water} = 25.180 \text{ g} - 20.175 \text{ g} = 5.005 \text{ g}$$

The density can be found by dividing the mass by the volume:

$$\text{Density} = \frac{\text{Mass}}{\text{Volume}} = \frac{5.005 \text{ g}}{5.0 \text{ ml}} = 1.0 \frac{\text{g}}{\text{ml}}$$

Note: According to rules for multiplication and division, we round the answer to two significant digits because the measurement with the fewest significant digits (5.0 mL) has two significant digits.

✓ Example 3.1

Using different volumes of water, experimental densities were found to be $0.98 \frac{\text{g}}{\text{ml}}$, $0.99 \frac{\text{g}}{\text{ml}}$, and $0.99 \frac{\text{g}}{\text{ml}}$. Calculate the average density and percent error.

Solution

To find the average, we add up each value and divide the sum by the total number.

$$\text{Average Density} = \frac{(0.98 + 0.99 + 0.99)}{3} = 0.99 \frac{\text{g}}{\text{ml}}$$

The percent error can be calculated using the following equation:

$$\text{Percent Error} = \frac{|E - T|}{|T|} \times 100$$

Where E is the experimental average density and T is the theoretical density value of water. This is an absolute value, meaning that even if the answer is negative, it is recorded as a positive value.

Using the average shown above and the true density of water, which is 1.0 g/mL, and multiplying the answer by 100, the percent error can be calculated as:

$$\text{Percent Error} = \frac{|0.99 - 1.0|}{|1.0|} \times 100 = 0.01 \times 100 = 1\%$$

The smaller the percent error, the closer the experimental value is to the true value.

✓ Example 3.1

A rectangular block having a mass of 45.250 g was shown to have a length of 2.75 cm, a width of 3.50 cm, and a height of 1.25 cm. Using this information, determine the density.

Solution

First, we need to find the volume of the rectangular block:

$$\text{Volume} = (2.75 \text{ cm}) \times (3.50 \text{ cm}) \times (1.25 \text{ cm}) = 12.0 \text{ cm}^3$$

To find the density, we divide the mass by the volume, as follows:

$$\text{Density} = \frac{45.250 \text{ g}}{12.0 \text{ cm}^3} = 3.77 \frac{\text{g}}{\text{cm}^3}$$

✓ Example 3.1

A rubber stopper weighing 10.130 g was added to 30.0 mL of water in a 100-mL graduated cylinder, raising the cylinder's volume to 36.5 mL. Using this information, calculate the density of the rubber stopper.

Solution

First, find the volume of the rubber stopper.

$$36.5 \text{ mL} - 30.0 \text{ mL} = 6.5 \text{ mL}$$

To find the density, we will divide the mass by volume.

$$\text{Density} = \frac{10.130 \text{ g}}{6.5 \text{ mL}} = 1.6 \frac{\text{g}}{\text{mL}} \text{ (or } 1.6 \frac{\text{g}}{\text{cm}^3}\text{)}$$

🖐 SAFETY PRECAUTIONS

1. Wear chemical splash goggles throughout the experiment.
2. Gloves are provided if you wish to wear them.
3. Dispose of all chemicals in the appropriate waste container (as indicated during the experimental procedure).
4. Be sure all glassware is clean and all equipment is returned to its proper place.
5. Clean the lab benches and check the laminated sheets to ensure all equipment is in your lab drawer before leaving the lab.
6. Remember, safety is paramount in the laboratory. Always wash your hands immediately upon leaving the lab to ensure you don't carry any chemicals.

EQUIPMENT *AND CHEMICALS NEEDED

Milligram balance

10 mL graduated cylinder

100 mL graduated cylinder

Centimeter ruler

Rectangular block

Rubber stopper

Deionized water

Cyclohexane

Amyl alcohol

Unknown liquid

* Images of equipment needed in this lab are in the appendix (the equipment may differ slightly or be subject to changes; follow your instructor's directions).

EXPERIMENTAL PROCEDURE

Part A: Sink or Float.

1. Add 2.0 mL of water and 2.0 mL of cyclohexane to a 10-mL graduated cylinder. Record your observations in the data sheet. Based on the table of density values (see part C), determine which liquid forms the top layer and which forms the bottom layer. Pour the contents of the graduated cylinder into the *nonhalogenated waste container*. Wash and dry the graduated cylinder before moving on to step 2.
2. Add 2.0 mL of water and 2.0 mL of amyl alcohol to a 10-mL graduated cylinder. Record your observations in the data sheet. Based on the table of density values (see part C), determine which liquid forms the top layer and which forms the bottom layer. Pour the contents of the graduated cylinder into the *nonhalogenated waste container*. Wash and dry the graduated cylinder before moving on to part B.

Part B: Density of Water

1. Using a milligram balance, determine the mass of an empty 10-mL graduated cylinder. Record this value in each of the three trials in the data table. (Remember, when recording mass from a milligram balance, write down the exact number being read by the balance.)
2. Add between 1.0 mL and 10.0 mL of deionized water to the graduated cylinder and write the exact volume of water used in trial one on the data sheet. (Remember to record volume from a graduated cylinder to the tenth decimal point.)
3. Obtain the combined mass of the graduated cylinder and the water. By subtracting the mass of the empty graduated cylinder from the combined mass, you can find the added mass of water. Record these values in trial 1 of the data sheet.
4. Calculate the density of water by dividing the mass by volume. Record your density value to 2 or 3 significant digits, depending on the amount of water you added. Write this value under trial 1 of the data sheet.
5. Repeat steps 2 - 4 two more times, using a different volume of water in each trial.
6. Calculate the average density of water by adding up the densities from the three trials and dividing that value by 3.

7. Calculate the percent error using the following formula: Percent Error = $\frac{|E-T|}{|T|} \times 100$

where E is the experimental average density calculated in step 6, and T is the theoretical density value of water. This is multiplied by 100 to get a percent. This is an absolute value, meaning that even if the answer is negative, it is recorded as a positive value.

Part C: Density of an Unknown Liquid

Part C: Table 3.1: Density of an Unknown Liquid

Liquid	Density
Water	1.00 g/mL
Methanol	0.791 g/mL
Ethanol	0.789 g/mL
Isopropyl alcohol	0.786 g/mL
Amyl alcohol	0.805 g/mL
Cyclohexane	0.779 g/mL
Methylene chloride	1.33 g/mL

Repeat part B (steps 1-6) using the unknown liquid. *Be sure the graduated cylinder is completely dry before beginning.* After each trial, pour the contents of the graduated cylinder into the *non-halogenated waste container*. Then, based on your average density and the densities in the table below, provide a possible identity for the unknown liquid.

Part D: Density of a Rectangular Block

1. Using a milligram balance, obtain the mass of a rectangular block. Be sure to record the block color used in the data table.
2. Using a centimeter ruler, measure the rectangular block's length, width, and height. Record your values to the hundredth of a decimal point.
3. Calculate the volume of the rectangular block to the appropriate number of significant digits using the following formula:
 $V = (\text{length}) \times (\text{width}) \times (\text{height})$

The volume will be expressed in units of cm^3 . (Note: $1 \text{ cm}^3 = 1 \text{ mL}$)

4. To find the density of the rectangular block, divide the mass by volume.

Part E: Density of a Rubber Stopper

1. Obtain the mass of a rubber stopper. Record the mass of both trials in the data table.

2. Fill a 100 mL graduated cylinder with 50.0 mL of water, and record this as the initial volume in trial 1.

3. Carefully add the rubber stopper to the water in the cylinder (avoid spilling out any water when the stopper is added). Record the new volume of water in the cylinder as the final volume in trial 1.

4. By subtracting the initial volume of water from the final volume of water, we can find the volume of the rubber stopper. (This is an application of Archimedes' Principle, which states that an object will displace a volume of water equal to its volume. We can use this principle to determine the volume of an irregularly shaped object, such as a rubber stopper.) Record the volume of the rubber stopper in trial 1.

5. To find the density of the rubber stopper, divide the mass by the volume, and record this value in trial 1 of the data table.

6. Repeat steps 2-5 using a different initial volume of water.

7. Calculate the average density of the rubber stopper.

PRE-LAB QUESTIONS

Name: _____

1. Explain how density can be used to infer the identity of an unknown.

2. An unknown liquid has a mass of 10.750 g and a volume of 11.5 mL. Calculate the density.

3. A rectangular block has a mass of 60.001 g. Using a centimeter ruler, the dimensions were determined as follows:

length = 2.30 cm

width = 5.60 cm

height = 1.11 cm

Using this information, determine the density of the rectangular block.

4. How can density be used to explain why solids float or sink when placed in liquids?

DATA AND OBSERVATIONS

Name: _____ Lab Partner(s): _____

Part A: Sink or Float.

1. When water and cyclohexane were mixed, which liquid formed the top layer?

2. When water and amyl alcohol were mixed, which liquid formed the top layer?

Part B: Density of Water

Part B: Table 3.1: Density of Water

	Trial 1	Trial 2	Trial 3
Mass of Empty Graduated Cylinder			
Mass of Graduated Cylinder and Water			
Mass of Water			
Volume of Water			
Density of Water			

Show calculations for the density of water for each trial:

Trial 1:

Trial 2:

Trial 3:

Average density of water (show calculation):

Percent error (show calculation):

Part C: Density of an Unknown Liquid

Unknown Liquid Number or Letter =

Part C: Table 3.1: Density of an Unknown Liquid

	Trial 1	Trial 2	Trial 3
Mass of Empty Graduated Cylinder			
Mass of Graduated Cylinder and Unknown			
Mass of Unknown			
Volume of Unknown			
Density of Unknown			

Show calculations for the density of the unknown liquid for each trial:

Trial 1:

Trial 2:

Trial 3:

Average density of unknown liquid (show calculations):

Possible identity of unknown liquid (from table):

Part D: Density of a Rectangular Block

Unknown rectangular block Number or Letter:

Color of Block:

Mass of rectangular block =

Length of rectangular block =

Width of rectangular block =

Height of rectangular block =

Volume of rectangular block (show calculations):

Density of rectangular block (show calculations):

Part E: Density of a Rubber Stopper

Part E: Table 3.1: Density of Rubber Stopper

	Trial 1	Trial 2
Mass of rubber stopper		
Initial volume of water		
Final volume of water		
Volume of rubber stopper		
Density of rubber stopper		

Show calculations for the density of the rubber stopper for each trial:

Trial 1:

Trial 2:

Average density of rubber stopper (show calculations):

POST LAB QUESTIONS

1. When 2.0 mL of water and 2.0 mL of an unknown liquid were added to a 10-mL graduated cylinder, water was shown to form the top layer. Using the table of densities in part C of the experiment, propose an identity for the unknown liquid.

2. List three potential sources of error for part B (density of water) and part C (density of unknown liquid) of this experiment.

3. In addition to density, what other properties could have been used to support the identity of the unknown liquid?

4. You are given a rectangular block that someone insists is gold. The block has a mass of 74.376 g and the following dimensions:

length = 1.50 cm; width = 2.75 cm; height = 2.01 cm

Part A: What is the density of the rectangular block?

Part B: Based on the density, could the block be gold? If not, what could it be? (Note: You will need to search for the density of gold.)

5. A rectangular sheet of aluminum foil, having a mass of 0.250 g, has a length of 5.75 cm and a width of 8.20 cm. If the density of aluminum is 2.70 g/cm^3 , find the aluminum foil's height (or thickness).

Please click [here](#) to access the Pre-Lab, Data Tables, and Post-Lab in Word or PDF format. Complete them and upload according to your instructor's instructions.

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4: LAB 4 - SEPARATION OF SALT AND SAND

PURPOSE

The purpose of this experiment is to:

- Separate a mixture containing an unknown composition of sand and salt.
- Calculate the percentage of sand and salt in the unknown mixture.

INTRODUCTION

Matter can be classified as a pure substance or a mixture. Pure substances include *elements* and *compounds*. Elements are the simplest forms of matter that cannot be decomposed into other types of matter. They are found on the periodic table. A compound is a chemical combination of elements. Mixtures can be either homogeneous, meaning they are uniform throughout, or heterogeneous, meaning they are non-uniform in composition.

Mixtures can be separated by physical methods such as filtration, decantation, distillation, and evaporation. In this experiment, you will use filtration and evaporation to separate a mixture of sand and salt. Once the mixture is separated, you will be asked to determine the percentage of sand and salt. The following example illustrates how to perform these calculations.

✓ Example 4.1

A mixture of sand and salt had a mass of 10.750 g. Once separated, the mass of sand was determined to be 5.550 g. The mass of recovered salt was

5.200 g. Using this information, determine the percentage of sand and salt in the mixture.

Solution

We find the percentage of a component in a mixture using the formula.

$$\frac{\text{Mass of Component}}{\text{Mass of mixture}} \times 100$$

$$\text{The percentage of sand would be } \frac{5.550 \text{ g}}{10.750 \text{ g}} \times 100 = 51.63 \%$$

$$\text{and the percentage of salt would be } \frac{5.200 \text{ g}}{10.750 \text{ g}} \times 100 = 48.37\%$$

SAFETY PRECAUTIONS

1. Always wear chemical splash goggles during the experiment.
2. Gloves are provided for those who wish to wear them.
3. Be cautious when handling glassware, both while it is being heated and after it has cooled, as it will remain hot. Allow all hot glassware to cool to room temperature before touching it.
4. At the end of the experiment, dispose of the sand in the “recovered sand jar” on the lab cart.
5. At the end of the experiment, wash the watch glass and the evaporating dish containing the recovered salt in the sink with plenty of soap and water.
6. Ensure all glassware is clean, and all equipment is returned to its designated place.
7. Clean the lab benches and verify all equipment is in your lab drawer before leaving.
8. Wash your hands as soon as you leave the lab.

EQUIPMENT* AND CHEMICALS NEEDED

Table 4.1: Equipment and Chemicals

Equipment	Equipment	Chemicals
Milligram balance	50.0 mL beaker	Filter paper
Spatula	Watch glass (2)	125 mL Erlenmeyer flask

Equipment	Equipment	Chemicals
Milligram balance	50.0 mL beaker	Deionized Water
Spatula	Watch glass (2)	Wash bottle
Bunsen burner	Glass funnel	125 mL Erlenmeyer flask
Matches, a striker, or a lighter	Wire gauze pad	Evaporating dish
Unknown mixture	Hot plate	Beaker tongs
Ring stand		Deionized Water
* Ring clamp	Stirring rod	Wash bottle
* Images of equipment needed in this lab can be found in the appendix (the equipment may differ or be subject to changes; follow your instructor's directions)		
Marbles or marbles (optional)	Glass funnel	Evaporating dish
Unknown mixture		

EXPERIMENTAL PROCEDURE

- Using a milligram balance, weigh an empty 50.0 mL beaker and record its Mass.
- Using a spatula, add approximately 1.000 - 2.000 g of an unknown mixture (record the mixture number or letter) to the 50.0 mL beaker.
- Obtain the mass of the mixture and the beaker and record the total mass. (Subtracting the mass of the empty beaker from the total mass will give you the mass of the mixture.)
- Using a wash bottle, add enough deionized water to the 50.0 mL beaker to completely cover the mixture. Continue stirring the mixture with a stirring rod for approximately five minutes to dissolve the salt.
- Using a glass funnel, a 125 mL Erlenmeyer flask, and filter paper, set up gravity filtration (following the instructions provided by your instructor) to separate the sand from the saltwater. A wash bottle can ensure the sand is completely removed from the beaker. Continue to rinse the beaker's contents into the filter until the beaker is empty.

To recover the sand, follow steps 6 through 9. To recover the salt, follow steps 10 through 14. These steps may be performed simultaneously depending on the number of members in your lab group.

- After filtration, remove the filter paper from the funnel. Using a wash bottle, wash the sand from the filter paper back into the 50 mL beaker, being sure to remove all the sand.
- Decant any excess water in the 50.0 mL beaker, then place the beaker on a hot plate with a low setting to dry the sand.

Note: Once the beaker warms, the sand may start to bubble. Hold the beaker with tongs and use a stirring rod to move the sand particles around it to prevent this.

- Once the sand is completely dry and does not clump together (it will appear like beach sand on a hot summer day), turn off the hot plate and allow the beaker to cool.
- Once the beaker is cool, obtain the mass of the 50.0 mL beaker and sand. (Subtracting the mass of the empty beaker from the combined mass of the beaker and sand will give you the mass of sand.)

- Obtain the mass of an empty evaporating dish and a watch glass.

- Pour the salt water from the 125 mL Erlenmeyer flask into the evaporating dish. Rinse the flask with deionized water from the wash bottle and pour this into the evaporating dish.

You can use a Bunsen Burner or a hot plate to evaporate the saltwater. See the steps below for either. Your instructor may decide which type of evaporation you need to use.

Using a hot plate:

- Place the Evaporating dish on the hot plate, drying the sand. Heat the mixture, and when the saltwater starts to boil, place a watch glass partially, and later entirely, on top of the evaporating dish to avoid splattering the saltwater into a crucible or beaker. Use a crucible or beaker tong for this purpose.

- Heat the mixture until all the water evaporates and there is no condensation on the watch glass. Then, turn off the hot plate, and allow the evaporating dish to cool.

Or

Using a Bunsen Burner:

12. Under the hood, set up a ring stand containing a ring clamp and wire gauze pad. Place a Bunsen burner a few inches below the wire gauze pad. **(Note: Your instructor will demonstrate how to set up this apparatus. Once you have set up your apparatus, ensure that your instructor approves it before proceeding with the next step.)**

13. Place the evaporating dish on the wire gauze pad and cover it with the watch glass. Light the Bunsen burner under the wire gauze pad and heat the mixture until all the water evaporates and there is no condensation on the watch glass. Then, turn off the burner and allow the evaporating dish to cool.

14. Once the evaporating dish is cool, weigh the evaporating dish, watch glass, and salt. (By subtracting the mass of the empty evaporating dish and the watch glass from the mass of the evaporating dish, watch glass, and salt, you will find the mass of salt.)

15. Clean up all equipment as specified in the Safety Precautions and Clean-Up section. Transfer all data to the assignment loaded in Blackboard and complete the lab calculations as instructed by your instructor.

PRE-LAB QUESTIONS

Name: _____

1. Imagine you are presented with a mixture of sand and salt. Based on previous knowledge, please describe how you would separate the mixture into its components.
2. Classify the following matter samples as a homogeneous mixture, a heterogeneous mixture, an element, or a compound.

Part A: red wine

Part B: carbon dioxide

Part C: pepperoni pizza

Part D: silver

Part E: brass

3. The Mass of a sand and salt mixture is 7.500 g. After separating the mixture, the mass of recovered sand was determined to be 5.473 g, and the mass of salt was 1.350 g. Calculate the percentage of sand and salt in the mixture.
4. Refer to pre-lab question 3. What is the total percent recovery of the components of the mixture? (Add the percent of sand and the percent of salt.)

DATA OBSERVATIONS AND CALCULATIONS

Name: _____ Lab Partner(s): _____

Unknown Letter / Number =

Table 4.2: Data table

--

Components	Mass (g) (Show calculations where possible)
1. Mass of an empty 50 mL beaker	
2. Mass of the beaker and the mixture	
3. Mass of mixture	
4. Mass of the beaker and sand	
5. Mass of sand	
6. Mass of empty evaporating dish and watch glass	
7. Mass of evaporating dish, watch glass, and salt.	
8. Mass of salt	

Percentage of sand in the mixture (show calculations):

Percentage of salt in the mixture (show calculations):

Percent recovery (show calculations):

POST LAB QUESTIONS

1. Explain why the percent recovery of your mixture may not have been equal to 100%.
2. Classify the substances encountered in this experiment as a homogeneous mixture, a heterogeneous mixture, an element, or a compound.

Deionized water:

Sand and salt mixture:

Sand:

Salt:

Saltwater:

3. Imagine you are provided with a mixture of ammonium chloride (which sublimates when heated), sand, and salt. Outline a procedure for separating the mixture into its components.
4. Chromatography is another separation method. What is chromatography, and how can it be used to separate the components of a mixture?

Please click [here](#) to access the Pre-Lab, Data Tables, and Post-Lab in Word or PDF format. Complete them and upload according to your instructor's instructions.

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5: LAB 5 - PHYSICAL AND CHEMICAL PROPERTIES

PURPOSE

The purpose of this experiment is to:

- Record the appearance of several metals and nonmetals.
- State whether a solid is soluble or insoluble in water.
- State whether a liquid is soluble or insoluble in water.
- Explain whether a substance is undergoing a physical or chemical change.
- Measure boiling points and record test tube reactions.

INTRODUCTION

Chemistry is the study of matter and any changes it undergoes. Matter is defined as anything that occupies space and has mass. Matter can be classified into three states: solid, liquid, and gas. These properties indicate that matter has specific physical characteristics, which correspond to the three **phases** of matter. A solid has a definite shape and volume. Solids are tightly packed, with atoms having a fixed volume and a definite shape. Liquids have a fixed volume but not a definite shape and take the shape of their containers. Gases have neither a definite shape nor a definite volume, and they expand to fill the space within their containers.

Matter can also be classified into mixtures and pure substances. A pure substance has a constant composition. All have the same composition and properties. Pure substances are divided into two classes: elements and compounds. Pure substances that cannot be broken down into simpler substances by chemical changes are called elements. Some examples are iron, silver, gold, aluminum, sulfur, oxygen, and copper. Pure substances that can be broken down to produce either elements or other compounds, or both, by chemical changes are called compounds.

Although there are 118 elements, tens of millions of chemical compounds result from different combinations of these elements. Each compound has a specific composition and definite chemical and physical properties that distinguish it from all other compounds. And, of course, there are innumerable ways to combine elements and compounds to form different mixtures.

Scientists use the periodic table to reference information about an element, such as atomic mass and chemical symbols. The periodic table's arrangement also allows scientists to discern trends in element properties, including electronegativity, ionization energy, and atomic radius. The periodic law states that the physical and chemical properties of the elements recur systematically and predictably when the elements are arranged in order of increasing atomic number.

The **periodic table** organizes elements based on their increasing atomic numbers, which allows for the prediction of their chemical and physical properties. Elements within the same group or column often share similar properties due to their similar outer electron configurations.

The properties that chemists use to describe matter fall into two general categories.

Physical properties are characteristics that describe matter. They include size, hardness, state, shape, color, density, melting, boiling, freezing, solubility, and mass.

Chemical properties describe how matter changes its chemical structure or composition. Flammability, toxicity, acidity, the reactivity of various types, and the heat of combustion

EQUIPMENT* AND CHEMICALS NEEDED

Table 5.1: Equipment and Chemicals

Equipment	Equipment	Chemicals
Safety goggles	Thermometer	1.0 M Hydrochloric Acid (HCl)
Spatula	Large test tube and split cork	Alcohol (ethanol/methanol)
Hot plate	Test tube holder	Sodium chloride (NaCl)
10 mL graduated cylinder	Medium-sized test tubes	Sodium carbonate (Na_2CO_3)
Glass Stirring Rod	Test tube rack	Calcium carbonate (CaCO_3)

Equipment	Equipment	Chemicals
250 ml beaker		Starch
Safety goggles	Thermometer	Hexane
Spatula	Large test tube and split cork	Alcohol (ethanol/methanol)
* Images of equipment needed in this lab can be found in the appendix (the equipment may differ or be subject to changes; follow your instructor's directions).	Test tube holder	Sodium chloride (NaCl)
Hot plate		
10 mL graduated cylinder	Medium-sized test tubes	Sodium carbonate (Na_2CO_3)
Glass Stirring Rod	Test tube rack	Calcium carbonate (CaCO_3)
1. Always wear chemical splash goggles during the experiment.		
2. 250 ml beaker provided for those who wish to wear them	Boiling chip	starch
3. Exercise extreme caution when using a hot plate or Bunsen burner to heat water.		Hexane

4. Water used for boiling point determination can be discarded in the sink. Other chemical waste solutions should be placed in appropriate waste containers as advised by the instructor.

5. Ensure all glassware is clean, and all equipment is returned to its designated place.

6. Thoroughly clean your work area and return chemicals and equipment to their designated places after the experiment.

7. Clean the lab benches and verify all equipment is in your lab drawer before leaving.

8. Wash your hands as soon as you leave the lab.

EXPERIMENTAL PROCEDURE

A. Identification, appearance, and observations of several metals, metalloids, and nonmetals.

Using the [virtual periodic table](#) and answering the questions in the data table.

1. Classify the following: Carbon, Calcium, Sodium, Nitrogen, Oxygen, Phosphorus, Neon, Vanadium, Lithium, as metals, metalloids, and nonmetals in the data sheet.
2. Identify the elements and group numbers of named groups: Alkali Metals, Alkaline earth metals, halogens, and noble gases.
3. Which group of elements are considered transition metals?
4. Which elements are considered Metalloids?

B. Physical and Chemical Properties and Changes

1. Take a 250 ml beaker and add approximately 175 ml of distilled water to it. Bring the mixture to a boil on a hot plate*.
2. While waiting for the water to boil, take a test tube large enough not to be immersed in the above 250ml beaker and add 25 drops of alcohol, along with a boiling chip. Note the alcohol's identity. Suspend the thermometer in the test tube using a split cork. Ensure the tip of the thermometer is at least 1 cm above the alcohol level in the test tube. Place the test tube in the test tube rack.
3. When the water starts boiling, record the boiling point in part B of the data table and observations. Identify whether this is a chemical or physical change. After recording the boiling point, turn off the hot plate.
4. Place the test tube with alcohol into the above beaker with hot water. Record the temperature as the alcohol starts to boil. If your water has cooled down and the alcohol does not boil, turn on your hot plate to warm the water.
5. After the remaining alcohol has cooled, pour it into the organic waste container, and the water can be discarded down the drain. Clean your work area and all glassware before moving onto the next step of the experiment.
6. Obtain six test tubes and place them on a test tube rack. Add 10 mL of distilled water to each. Add half a spatula or 25 drops of the mentioned chemicals to the test tubes. Into the Test tube: 1 Sodium chloride (NaCl), 2 Sodium carbonate (Na_2CO_3), 3 Calcium carbonate (CaCO_3), 4 Starch, 5 twenty five drops of ethanol (or methanol), 6 25 drops of hexane, respectively (add only one compound into each test tube of distilled water). Swirl the liquid in the test tubes, and if necessary, use the glass rod to stir the solid in all three test tubes. Record the observations and solubilities and identify the type of change involved. Chemical waste solutions should be placed in appropriate waste containers.

7. Obtain four test tubes, place them on a test tube rack, and add 10 mL of 1 M HCl to each. In each test tube, add a small amount, around half a spatula each of Sodium carbonate Na_2CO_3 , Calcium carbonate CaCO_3 , 25 drops of ethanol (or methanol), and 25 drops of hexane, respectively. Stir using the glass rod. (Remember to clean the glass rod after mixing it into each test tube.) Record the observations and solubilities and identify the type of change involved.
8. As instructed by your instructor, discard the contents of the test tubes into the designated waste container. Thoroughly clean all glassware with soap and water. Return all equipment and chemicals to their appropriate place. Before leaving the lab, clean the bench and your work area.

PRE-LAB QUESTIONS

Name _____

- How are the elements in the periodic table arranged? Why is it important?
- The columns and rows of the periodic table are called:
- The elements in the two rows at the bottom of the periodic table are called:
- Define what a Physical property is and give five examples.
- Define Chemical property and give five examples.

DATA AND OBSERVATIONS

Name: _____ Lab Partner(s): _____

A. Identification, appearance, and observations of several metals, metalloids, and nonmetals.

- Classify the following as metals, metalloids, and nonmetals.

Table 5.2: Classify the following as metals, metalloids, and nonmetals.

Element Name	Symbol of Element	Identify whether Metal / Nonmetal / Metalloid	Group #	State (s, l, g)
Carbon				
Calcium				
Sodium				
Nitrogen				
Oxygen				
Phosphorous				
Neon				
Vanadium				
Lithium				

Element Name	Symbol of Element	Identify whether Metal / Nonmetal / Metalloid	Group #	State (s, l, g)
Germanium				

2. Identify the elements and group numbers of named groups: Alkali Metals, Alkaline earth metals, halogens, and noble gases.

a) **Alkali Metals**: Group Number

Names and Symbols of Alkali Metals:

b) **Alkaline Earth Metals** Group Number:

Names and Symbols Alkaline Earth Metals:

c) **Halogens** Group Number:

Names and Symbols of Halogens:

d) **Noble Gases** Group Number:

Names and Symbols of Noble Gases:

3. **Transition metals**, Group Numbers:

Names and Symbols of five Transition metals:

9. **Names and Symbols of Metalloids**:

B. Physical and Chemical Properties and Changes

1. Boiling Point:

i) Boiling Point of Water:

Identify whether the boiling point of water is a Physical or Chemical change:

ii) Boiling Point Alcohol:

Identify whether the boiling point of Alcohol is a Physical or Chemical change:

2. Determine whether a change is chemical or physical using Water.

Table 5.3: Classify Chemical or Physical Change using Water

Test tubes #	Observation	Soluble / Insoluble	Physical /Chemical Change
1. Sodium chloride + water (NaCl + H ₂ O)			
2. Sodium Carbonate + water (Na ₂ CO ₃ + H ₂ O)			
3. Calcium Carbonate + water (CaCO ₃ + H ₂ O)			
4. Starch + water			

Test tubes #	Observation	Soluble / Insoluble	Physical /Chemical Change
5. 25 drops of ethanol + water or (25 drops of methanol + water)			
6. 25 drops of hexane + water			

3. Identify if a Chemical or Physical change using Hydrochloric acid.

Table 5.4: Classify Chemical or Physical change using Hydrochloric acid

Test tubes #	Observation	Soluble / Insoluble	Physical /Chemical Change
1. Sodium chloride + HCl (aq) (NaCl + HCl (aq))			
2. Sodium Carbonate + HCl (aq) Na_2CO_3 + HCl (aq)			
3. Calcium Carbonate + HCl (aq) CaCO_3 + HCl (aq)			
4. Starch+ HCl (aq)			
5. 25 drops of ethanol + HCl (aq) (or methanol + HCl (aq))			
6. 25 drops of hexane + HCl (aq)			

POST-LAB QUESTIONS

1. Classify the following elements and complete the table below.

Table 5.5: Classify the Elements

Name of Element	Symbol of Element	Identify whether Metal / Nonmetal/Metalloid	Group number	Period
Lithium				
	Si			
Zinc				
	Mg			
Copper				
	I			
Uranium				
	Cl			
Tin				
	Ge			

2. Identify whether a chemical or physical change.

Table 5.6: Classify the Elements

--

	Chemical or Physical Change
1. The density of zinc is 7.41 g/cm ³	
2. Zinc combines with Hydrochloric acid to give zinc chloride and hydrogen gas	
3. Silver tarnishes to silver oxide when exposed to moisture	
4. Melting of polar ice caps due to global warming	
5. Copper can be drawn into a thin wire and wrapped into a cord	
6. An iron nail rusts when exposed to moisture	

Please click [here](#) to access the Pre-Lab, Data Tables, and Post-Lab in Word or PDF format. Complete them and upload according to your instructor's instructions.

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6: LAB 6 - CHEMICAL REACTIONS AND EQUATIONS

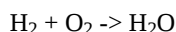
PURPOSE

The purpose of this experiment is to:

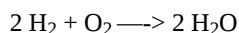
- Run several chemical reactions and record observations.
- Balance chemical equations.
- Predict the products of single and double replacement reactions.

INTRODUCTION

Chemical changes produce new materials. In a chemical change, atoms are rearranged as reactants (the starting materials) and are converted into products (the new materials). Chemical equations can represent chemical changes. Consider the conversion of hydrogen and oxygen into water:

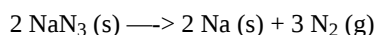


In a chemical reaction, matter cannot be created or destroyed, meaning the same number and type of atoms must be shown on each side of the equation. To illustrate this, we must balance the equation by adding a whole number coefficient in front of each element or compound. We can balance the above equation as follows:

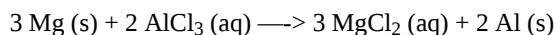


The total number of atoms on each side of the equation can be found by multiplying the added coefficient by the given subscript. On the left side of the equation, we see four hydrogens (2×2) and two oxygens (1×2). The right side of the equation also has four hydrogens (2×2) and two oxygens (2×1). Since this equation has the same number of atoms on each side, it is balanced. Refer to the course lectures and notes for examples and detailed steps on balancing equations.

In this experiment, you will explore three reaction types: decomposition, single, and double displacement. A decomposition reaction occurs when one reactant forms two or more products. An example would be the conversion of sodium azide into sodium metal and nitrogen gas, as shown below:

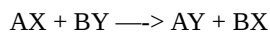


In a single displacement reaction, one element replaces another in a compound. These reactions often run in aqueous solutions. Consider the reaction of magnesium metal with aluminum chloride:

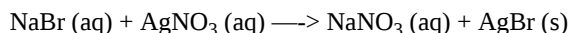


In this reaction, magnesium replaces aluminum with aluminum chloride.

Like single displacement reactions, double displacement reactions also occur in aqueous solutions, and cations and anions exchange places according to the following generic equation:



In many cases, when two aqueous solutions are mixed, a solid, known as a precipitate, forms, as shown by the reaction between sodium bromide and silver nitrate:



To predict the products of a double displacement reaction, write the four ions present in the solution (in this case, Na^+ , Br^- , Ag^+ , and NO_3^-) and pair opposite ions together, writing the formula for the ionic compound. One quick way to identify a precipitate (if it forms) is by looking at the two cations. The precipitate typically forms from the heavier metal cation. In this reaction, silver is heavier than sodium. A more formal method for determining whether a precipitate will form when two aqueous solutions are mixed is by examining a list of solubility guidelines provided in the course textbook.

SAFETY PRECAUTIONS

1. Always wear chemical splash goggles while working on this experiment.

2. Wearing gloves, especially when working with silver nitrate, is advisable, as it can cause skin stains. Be sure to thoroughly wash your hands after completing this experiment.
3. Exercise caution when working with Bunsen burners. Keep notebooks, purses, books, and other equipment away from open flames. Follow all directions provided by your instructor when heating test tubes.
4. Thoroughly read and reread the bottles' labels to ensure you mix the appropriate solutions.
5. Dispose of chemical waste in the designated waste containers, as instructed by your instructor.
6. Thoroughly clean your work area and return chemicals and equipment to their designated places after the experiment.

EQUIPMENT* AND CHEMICALS NEEDED

Bunsen burner with hose

Well plate

Spatula

Wire cutters or scissors

600 mL beaker

Test tube holder

Test tube rack

5 medium-sized test tubes

Matches, lighter, or Striker.

10 mL graduated cylinder

copper wire

Solid NaHCO_3

Solid $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$

Solid magnesium and zinc

1 M HCl

Solid waste container

Inorganic waste container

0.5-1 M solutions of CuSO_4 , AgNO_3 , NaCl , CaCl_2 , Na_3PO_4 , BaCl_2 , Na_2SO_4 , NaOH , FeCl_3

* Images of equipment needed in this lab can be found in the appendix (the equipment may differ or be subject to changes; follow your instructor's directions).

EXPERIMENTAL PROCEDURE

Part A: Decomposition Reactions

1. Gather the following equipment and assemble the Bunsen burner.

Bunsen burner with the hose, matches or lighter, test tube holder, test tube rack, and two medium-sized test tubes.

2. Take two medium-sized test tubes. Add a small scoop of solid NaHCO_3 (baking soda) to the first medium-sized test tube and a small scoop of solid $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ to the second medium-sized test tube. Note: Rinse and dry your spatula between obtaining the two solids to avoid cross-contamination.
3. Using a test tube holder, place the tip of the test tube containing NaHCO_3 into the flame for a few minutes. Hold the test tube at an angle, ensuring the open end is not pointed towards you or your lab partners. Record any observations.
4. Using a test tube holder, place the tip of the test tube containing $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ into the flame for a few minutes. Hold the test tube at an angle, ensuring the open end is not pointed towards you or your lab partners. Record any observations.

4. Turn off the burner. After the remaining solids in the test tubes have cooled, pour them into the inorganic waste container. Clean your work area and all glassware before proceeding to the next step of the experiment.

Part B: Single Displacement Reactions

1. Take five test tubes, add a small scoop of Mg (s), Zn (s), and Cu (s) to three separate test tubes, a small scoop of Zn (s) to the fourth test tube, and a small piece of copper wire to the fifth test tube. Secure the test tubes in the test tube rack.
2. Add approximately 2.0 mL of 1.0 M HCl into the **FIRST THREE test tubes ONLY** and record your observations.
3. Into the **fourth** test tube with a small scoop of Zn (s), add enough CuSO₄ solution to cover the Zn (s) completely. Record your initial observation and then your observations after 15 minutes and 30 minutes. (While waiting, you can work on Part C)
4. Using a small piece of copper wire, add enough AgNO₃ solution to completely cover the Cu (s) in the fifth test tube. Record your initial observation and then your observations after 15 and 30 minutes.
5. Leave the test tubes in the rack, and after the 30-minute observation, pour out the contents of the test tubes into a 600 mL beaker. Using forceps, remove any unreacted metal and place it on a paper towel. Place the solution into the inorganic waste container and dry the unreacted metal in the metal waste. Clean and dry the test tubes.

Part C: Double Displacement Reactions

1. Obtain a well plate, and add 5-10 drops of each solution to the following wells:

Well 1: NaCl and AgNO₃

Well 2: CaCl₂ and Na₃PO₄

Well 3: BaCl₂ and Na₂SO₄

Well 4: CuSO₄ and Na₃PO₄

Well 5: CuSO₄ and NaOH

Well 6: FeCl₃ and NaOH

Well 7: CaCl₂ and Na₂SO₄

Well 8: FeCl₃ and Na₂SO₄

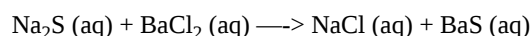
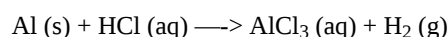
2. Record your observations. Pour the contents of the well plate into a 600.0 mL beaker. The contents of the 600 mL beaker should then be poured into the inorganic waste container.
3. Thoroughly clean all glassware with soap and water. Return all equipment and chemicals to their appropriate places. Clean your work area before leaving the lab.

PRE-LAB QUESTIONS

Name: _____

1. Provide three examples of evidence showing that a chemical reaction has occurred.

2. Balance the following chemical equations:



3. Using a list of solubility guidelines, predict whether the following compounds will be soluble or insoluble in water.

Pb(NO₃)₂:

PbCl₂:

SrS:

Fe(OH)₃:

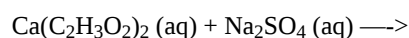
Na₂SO₄:

AlPO₄:

CuCO₃:

KI:

4. Predict the products and their physical states for each reaction. (Hint: Refer to a table of solubility guidelines to determine whether a precipitate would form.). Then, balance the overall equation.



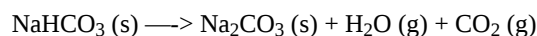
DATA AND OBSERVATIONS

Name: _____ Lab Partner(s): _____

Part A: Decomposition Reactions

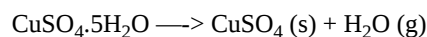
For each of the reactions, **balance** the provided equation. Record all **observations and evidence of reaction**.

Reaction 1: NaHCO₃ and heat



Observations:

Reaction 2: CuSO₄ · 5H₂O and heat



Observations:

Part B: Single Displacement Reactions

Predict the products' physical states for each reaction, and **balance the equation**. If no reaction occurs, write NR. Record all **observations and evidence of response**.

Reaction 1: Mg and HCl



Observations:

Reaction 2: Zn and HCl



Observations:

Reaction 3: Cu and HCl



Observations:

Reaction 4: Zn and CuSO₄



Observations:

Reaction 5: Cu and AgNO₃



Observations:

Part C: Double Displacement Reactions

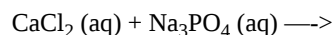
Predict the physical states of the products for each reaction and **balance** the equation. If no reaction occurs, write NR. Record all **observations and evidence of response**.

Reaction 1: NaCl and AgNO₃



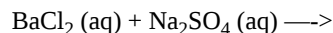
Observations:

Reaction 2: CaCl₂ and Na₃PO₄



Observations:

Reaction 3: BaCl₂ and Na₂SO₄



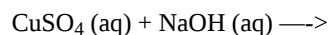
Observations:

Reaction 4: CuSO₄ and Na₃PO₄



Observations:

Reaction 5: CuSO₄ and NaOH



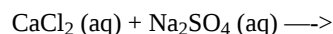
Observations:

Reaction 6: FeCl₃ and NaOH



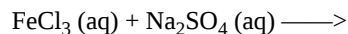
Observations:

Reaction 7: CaCl₂ and Na₂SO₄



Observations:

Reaction 8: FeCl₃ and Na₂SO₄



Observations:

POST LAB QUESTIONS

1. Another type of reaction is the redox (or oxidation-reduction). Which of the three reactions (decomposition, single displacement, or double displacement) explored during this lab can also be classified as redox reactions? Explain.
2. An activity series can be used to predict whether a single displacement reaction will occur as written. Any metal on the list can react when placed into solutions containing ions listed below it in the activity series. Access an activity series (from either the course textbook or by doing a Google search) and briefly explain how it summarizes the results of the reactions observed in part B.
3. For part C, did any reactions not produce a precipitate? If yes, which one(s)? Would these still be classified as chemical changes? Explain.

Please click here to access the Pre-Lab, Data Tables, and Post-Lab in Word or PDF format. Complete them and upload according to your instructor's instructions.

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7: LAB 7 - FLAME TESTS AND IDENTIFICATION OF AN UNKNOWN

PURPOSE

The purpose of this experiment is to:

- To perform flame tests on solutions containing various elements and examine the colors produced.
- Record observations from several double displacement reactions.
- Identify an unknown based on the results of flame tests and double displacement reactions.

INTRODUCTION

In the lecture, you learned about two models of atomic structure: the Rutherford and Bohr models. The Rutherford model states that most of the atom's mass comes from the nucleus, which is positively charged and resides at the center of the atom. In this model, electrons randomly orbit the nucleus in circular paths. The Bohr model explains that protons and neutrons are inside the atom's nucleus. Electrons exist at specific energy levels outside the nucleus.

When solutions containing various elements are heated, electrons transition from a lower to a higher energy level. Electrons eventually will return to their original energy levels, and when they do so, they often emit energy in the form of visible light. Evidence of the Bohr model is observed by the light produced as materials containing certain elements are heated. Because electrons exist at specific energy levels and are not randomly placed outside the nucleus, each component has its characteristic pattern of energy levels, and, as a result, emits its characteristic pattern of light.

In this experiment, you will run flame tests on several known solutions and an unknown. You will react to these same chemical solutions with various reagents and record the observations of the resulting double displacement reactions. Based on these results, you can identify the components of an unknown solution.

SAFETY PRECAUTIONS

1. Always wear chemical splash goggles when working in the lab.
2. Do not place all materials (book bags, purses, notebooks, etc.) near open flames.
3. Exercise caution when working with Bunsen burners and have your instructor check your setup before lighting the burner.
4. Turn off the gas immediately after working with the Bunsen burner.
5. Carefully handle all solutions used in this experiment. Use the gloves provided.
6. Place all materials in the appropriate waste container and clean your work area when finished. Wash your hands once you leave the lab.

EQUIPMENT* AND CHEMICALS NEEDED

Table 7.1: Equipment and Chemicals

Equipment	Chemicals
Well plate	Dilute solution of AgNO_3
Nichrome wire	1 M HCl
Bunsen burner with hose	Dilute solutions (~0.5 M) of either the chlorides or nitrates of Li, Na, K, Ca, Sr, Ba, and Cu^{+2}
50 mL beaker	Dilute solutions of Na_2CO_3 , Na_3PO_4 , and Na_2SO_4
600 mL beaker	Unknown solutions containing the chlorides, bromides, or iodides of the metal cations listed above
light source, Matches, lighter, or Striker.	Dilute solutions of NaCl , NaBr , and NaI

* Images of equipment needed in this lab can be found in the appendix (the equipment may differ or be subject to changes; follow your instructor's directions).

EXPERIMENTAL PROCEDURE

Part I: Flame Tests

1. Add 10 drops of solutions containing Li, Na, K, Ca, Sr, Ba, Cu^{+2} , and an unknown to individual wells in a well plate. Be sure to record the unknown letter or number.
2. Using a 50 mL beaker, obtain 10.0 mL of 1.0 M HCl.
3. Obtain a Bunsen burner with a hose and nichrome wire.
4. Bring all the equipment (listed in 1-3) to your lab bench. Once you are ready to begin, carefully turn on/light the burner, as guided by your instructor.
5. Dip the nichrome wire into the HCl solution and hold the tip in the flame until it produces no color. This amounts to cleaning the wire.
6. Dip the tip of the wire into the lithium solution and hold the wire in the flame until a specific color is produced. You may repeat this procedure several times to saturate the wire with the lithium solution.
7. Once you observe the color lithium produces in the flame, repeat steps 5-6 for the remaining solutions.
8. When you finish this part of the experiment, clean the nichrome wire and turn off the gas thoroughly.
9. Pour the contents of the well plate and 50 mL beaker into a 600 mL beaker and transfer the contents from the 600 mL beaker into the inorganic waste container. Be sure to clean and dry all equipment before proceeding to part II.

Part II: Double Replacement Reactions

1. Add 10 drops of solutions containing Li, Na, K, Ca, Sr, Ba, Cu^{+2} , and an unknown to individual wells in a well plate. Use the same unknown as you used for part I.
2. Add a few drops of Na_2SO_4 solution to each well, and record your observations in the data table. If no reaction occurs, write NR. Otherwise, observations can be written as white solid, bubbles formed, etc.
3. Pour the contents of the well plate into a 600 mL beaker and transfer the contents from the 600 mL beaker into the inorganic waste container. Be sure to clean and dry all equipment.
4. Repeat steps 1-3 using Na_3PO_4 instead of Na_2SO_4 .
5. Repeat steps 1-3, using Na_2CO_3 instead of Na_2SO_4 and Na_3PO_4 .
6. Based on Part I and II results, you should now be able to identify the cation in your unknown solution.

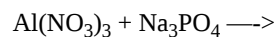
Part III. Identification of Unknown Anion

1. Add 10 drops of NaCl, NaBr, NaI, and the same unknown you used for parts I and II to individual wells in a well plate.
2. Add a few drops of AgNO_3 solution to each well, and record your observations.
3. Pour the contents of the well plate into a 600 mL beaker and transfer the contents from the 600 mL beaker into the inorganic waste container. Be sure to clean and dry all equipment.
4. Based on these results, you should now be able to identify the anion in your unknown solution.

PRE-LAB QUESTIONS

Name: _____

1. Briefly explain how the Bohr model can be applied to colors observed from fireworks.
2. What is an electronic configuration? Which model (Rutherford or Bohr) supports the concept of electronic configurations?
3. Determine the products of the following double displacement reactions occurring in aqueous solution and balance the resulting equation. (Note: You need not record the physical states of the substances.)



DATA AND OBSERVATIONS

Name: _____ Lab Partner(s): _____

Unknown Letter / Number =

Part I: Flame Tests

Table 7.2: Part I Flame Tests

Solution	Color
Lithium	
Sodium	
Potassium	
Calcium	
Strontium	
Barium	
Copper	
Unknown	

Part II: Double Replacement Reactions

Table 7.3: Part II Double Replacement Reactions

Solution	Reaction with Na_2SO_4	Reaction with Na_3PO_4	Reaction with Na_2CO_3
Lithium			
Sodium			
Potassium			
Calcium			
Strontium			
Barium			
Copper			
Unknown			

Part III: Identification of Unknown Anion

Table 7.4: Part III Identification of Unknown Anion

Solution	Results
NaCl	
NaBr	

Solution	Results
NaI	
Unknown	

Based on the results from parts I and II, the **cation** in the unknown solution is: _____.

Based on the results from part III, the **anion** in the unknown solution is _____.

POST-LAB QUESTIONS

1. Write electronic configurations for the following elements used in this experiment: Li, Na, K, Ca, Sr, Ba.
2. Refer to post-lab question 1. Which elements have similar electronic configurations? What is the implication of this?
3. Refer to part II of the data and observations section. Write the balanced chemical equation for each reaction that produces a solid, showing the physical states of all reactants and products.
4. Many streetlights are orange/yellow. Based on the results from part I, what element may be present in streetlights?
5. Based on the results from part I, which element(s) may be present in colors produced by the following fireworks?

A: Violet

B: Crimson red

Please click here to access the Pre-Lab, Data Tables, and Post-Lab in Word or PDF format. Complete them and upload according to your instructor's instructions.

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8: LAB 8 - EMPIRICAL FORMULAS AND HYDRATES

PURPOSE

The purpose of this experiment is to:

- Determine the empirical formula of a hydrate.
- Calculate the percentage of water in a hydrate.

INTRODUCTION

An empirical formula is the smallest whole-number ratio of elements in a compound. It can be determined experimentally by examining a substance's molar ratio of individual components. Consider the following example:

✓ Example 8.1

A compound was shown to consist of 0.750 g of carbon and 2.000 g of oxygen. What is the empirical formula?

Solution

To calculate the empirical formula, we would first need to convert grams of each element to moles, as shown below:

$$0.750 \text{ g-C} \times \frac{1 \text{ mol C}}{12 \text{ g-C}} = 0.0625 \text{ mol C}$$

$$2.000 \text{ g-O} \times \frac{1 \text{ mol O}}{16 \text{ g-O}} = 0.125 \text{ mol O}$$

Since we know the ratio of moles, we can automatically determine the ratio of atoms, as 1 mole equals 6.02×10^{23} atoms. Therefore, the ratio of moles of the carbon oxide compound is:

$$\frac{0.125 \text{ mol}}{0.0625 \text{ mol}} = 2$$

Since there are twice as many moles of oxygen as carbon, the empirical formula is CO_2 .

In this experiment, you will determine the empirical formula of a hydrate, a compound containing water molecules. First, you will heat the hydrate to drive off the water molecules and then determine the mass of water present in the hydrate by subtracting the mass of the anhydrous compound from the mass of the original hydrate. The empirical formula will be found by comparing the moles of anhydrous compound to the moles of water, as shown in the example below:

✓ Example 8.1

2.500 g of a copper(II) chloride hydrate was heated, resulting in 1.972 g of anhydrous compound. Using this information, determine the empirical formula of the copper chloride hydrate.

Solution:

First, we can find the mass of water driven off:

$$2.500 \text{ g hydrate} - 1.972 \text{ g anhydrous compound} = 0.528 \text{ g water}$$

Next, we can find the molar ratio of the anhydrous compound (AC) to water. Since the anhydrous compound has the formula CuCl_2 , we can convert grams to moles, as follows:

$$1.972 \text{ g AC} \times \frac{1 \text{ mol AC}}{134.45 \text{ g AC}} = 0.0147 \text{ mol AC}$$

Similarly, we can find the moles of water:

$$0.528 \text{ g-water} \times \frac{1 \text{ mol water}}{18 \text{ g-water}} = 0.0293 \text{ mol water}$$

Comparing the molar ratios gives:

$$\frac{0.0293 \text{ mol}}{0.0147 \text{ mol}} = 2$$

The empirical formula would therefore be written as $\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$, indicating that there are two water molecules for every molecule of the anhydrous compound. Since the molar mass of the hydrate is 170.45 g/mol (adding up the molar masses from Cu, 2 Cl, and 2 H_2O), the percentage of water can be calculated as:

$$\% \text{ water} = \left(\frac{\text{mass of water}}{\text{mass of hydrate}} \right) \times 100 = \left(\frac{0.528 \text{ g}}{2.500 \text{ g}} \right) \times 100 = 21.12\%$$



SAFETY PRECAUTIONS

1. Always wear chemical splash goggles when working in this lab.
2. Gloves are provided, if you choose to use them.
3. Be sure all notebooks, book bags, purses, etc., are not placed near open flames.
4. Be careful when handling hot glassware.
5. Extinguish flames when not in use.
6. Clean your work area, and place all waste/equipment in the appropriate place when finished.
7. Wash your hands once you leave the lab.

EQUIPMENT* AND CHEMICALS NEEDED

Table 8.1: Equipment and Chemicals

Equipment	Equipment and Chemicals
Milligram balance	Watch glass
spatula	250 mL beaker
Ring stand	Matches, lighter, or Striker
Ring clamp	Beaker tongs
Wire gauze/gauze pad	Bunsen Burner and tubing
	Unknown- Alum (Potassium aluminum sulfate hydrate)

* Images of equipment needed in this lab are in the appendix (the equipment may differ slightly or be subject to changes; follow your instructors' directions).

EXPERIMENTAL PROCEDURE

- 1) Obtain the combined mass of an empty 250.0 mL beaker and a watch glass.
- 2) Add approximately 1.000 g – 1.500 g of hydrate (formula of the anhydrous compound is $\text{KAl}(\text{SO}_4)_2$) to the beaker, and get the combined mass of the beaker, watch glass, and hydrate.
- 3) Set up a ring stand with a ring clamp and a wire gauze/gauze pad. Place a Bunsen burner a few inches below the gauze pad. Once your setup is complete, have your instructor check it before proceeding to the next step.



- 4) Place the beaker containing the hydrate on the gauze pad and cover it with the watch glass. Turn on the burner (following all safety procedures presented by your instructor) and heat the hydrate. You will notice moisture forming on the beaker and the watch glass. Heat the hydrate for an additional two minutes, until no moisture remains.
- 5) Extinguish the burner, and allow the beaker to cool to room temperature. It may take up to 20 minutes to cool to room temperature..
- 6) Once the beaker is cool, get the combined mass of the beaker, watch glass, and anhydrous compound.
- 7) Dispose of the remaining solid in the solid inorganic waste container, wash and dry your glassware, and complete steps 1-7 for a second trial.

PRE-LAB QUESTIONS

Name: _____

1) In your own words, define the following:

Empirical formula:

Hydrate:

Anhydrous compound:

2) 1.000 g of a magnesium sulfate (MgSO_4) hydrate was heated, forming 0.489 g of anhydrous compound. Using this information, determine the empirical formula of the hydrate and the percentage of water present in the hydrate.

DATA AND OBSERVATIONS

Name: _____ Lab Partner(s): _____

Table 8.2: Data Table

Data	Trial 1	Trial 2
Empirical formula of hydrate		
Mass of anhydrous compound (M_{AC}) ($M_{AC} = M_3 - M_1$)		
Mass of beaker and watch glass (M_1)		
Mass of beaker, watch glass, and anhydrous compound (M_3) (After heating)		
Mass of beaker, watch glass, and hydrate (M_2)		
Mass of hydrate (M) ($M = M_2 - M_1$)		
Mass of water driven off (M_{H_2O}) ($M_{H_2O} = M - M_{AC}$)		
Moles of anhydrous compound, $\text{KAl}(\text{SO}_4)_2$		
Moles of water		

Show all of your calculations here for both trials:

POST-LAB QUESTIONS

1. List some potential sources of error in determining the empirical formula of the hydrate.
2. What is the average percentage of water in the $\text{KAl}(\text{SO}_4)_2$ hydrate? (Show calculation.)
3. A hydrate of $\text{Na}_3\text{B}_4\text{O}_7$ contained 62.23% water by mass. Using this information, determine the empirical formula of the hydrate. (Hint: Assume 100 g of compound.)

Please click [here](#) to access the Pre-Lab, Data Tables, and Post-Lab in Word or PDF format. Complete them and upload according to your instructor's instructions.

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9: LAB 9 - DECOMPOSITION OF BAKING SODA

PURPOSE

The purpose of this experiment is to:

- Demonstrate decomposition of a compound.
- To experimentally determine the stoichiometry of the thermal decomposition of baking soda, (NaHCO_3).
- Estimate the percentage yield of sodium carbonate from a decomposition reaction.

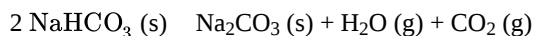
INTRODUCTION

Stoichiometry is based on the law of conservation of mass, which states that the total mass of a reactant is equal to the total mass of products, where relationships among the quantities of reactants and products typically form a mole ratio, stoichiometric coefficients. Stoichiometric coefficients are also helpful in determining the mole ratio between the reactants and products. The mole ratio is important because it enables chemists to calculate the number of moles of product formed from a given number of reactants, or the number of moles required to produce a certain product.

When two reactants are mixed, one may be used up completely, while the other may remain in excess. The amount of product formed will be limited by the reactant that is used up first. A balanced chemical equation helps students determine the amount of product formed (actual yield), which reactant runs out first (limiting reagent), and which remains (excess reagent).

The thermal decomposition (decomposition by heating) of sodium bicarbonate (NaHCO_3) is a common chemical reaction. Sodium bicarbonate, more commonly known as baking soda, is widely listed as an ingredient in baked goods recipes. As the food item is being cooked or baked, the baking soda undergoes decomposition, releasing gas and causing the food item to “rise” and have a “light” texture.

In this experiment, we will use stoichiometry to determine the percent yield of sodium carbonate (Na_2CO_3) formed. When heated, baking soda, NaHCO_3 (sodium bicarbonate or sodium hydrogen carbonate) decomposes, yielding Sodium carbonate (Na_2CO_3), water (H_2O), and carbon dioxide (CO_2).



Decomposition by heating is called thermal decomposition. During the thermal decomposition of baking soda, NaHCO_3 , the products water (H_2O) and Carbon dioxide (CO_2) are released as gases, and the sodium carbonate is left as a solid. The amount (mass) of (Na_2CO_3) obtained is the actual yield.

The theoretical yield, the maximum amount of (Na_2CO_3) that could be produced under the given conditions, is calculated using the mass of baking soda used and mole ratios from the balanced chemical equation, as shown in Example 1.

Percent Yield of Sodium Carbonate, (Na_2CO_3), obtained by decomposition of Baking Soda, NaHCO_3

Example 1:

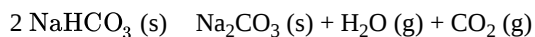
A student heated 3.255 g of baking soda (Sodium bicarbonate), NaHCO_3 , in a lab to produce 1.875 g of sodium carbonate, Na_2CO_3 . Calculate the percent yield.

Solution:

$$\text{Percent Yield} = \frac{\text{Actual Yield}}{\text{Theoretical Yield}} \times 100$$

To find the percent yield, we need to calculate the theoretical yield, as the actual yield is given as the mass of product obtained after the chemical reaction.

The theoretical yield in this case is the amount of Na_2CO_3 produced from 3.255 g of baking soda, using the molar masses of baking soda and sodium carbonate and the coefficients of the balanced chemical equation as a mole ratio.



g NaHCO_3 mol NaHCO_3 mol Na_2CO_3 g Na_2CO_3

Text Box: Molar mass of $\text{NaHCO}_3 = (1 \times \text{Na}) + (1 \times \text{H}) + (1 \times \text{C}) + (3 \times \text{O}) = (1 \times 22.99 \text{ g}) + (1 \times 1.01 \text{ g}) + (1 \times 12.01 \text{ g}) + (3 \times 16.00 \text{ g}) = 84.01 \text{ g}$ Molar mass of $\text{Na}_2\text{CO}_3 = (2 \times \text{Na}) + (1 \times \text{C}) + (3 \times \text{O}) = (2 \times 22.99 \text{ g}) + (1 \times 12.01 \text{ g}) + (3 \times 16.00 \text{ g}) = 105.99 \text{ g}$

The balanced chemical equation indicates that two moles of NaHCO_3 yield one mole of Na_2CO_3 . Using the molar masses of NaHCO_3 and Na_2CO_3 , we can calculate the theoretical yield.

Theoretical yield = $3.255 \text{ g NaHCO}_3 \times \frac{1 \text{ mol NaHCO}_3}{84.01 \text{ g NaHCO}_3} \times \frac{1 \text{ mol Na}_2\text{CO}_3}{2 \text{ mol NaHCO}_3} \times 105.99 \text{ g Na}_2\text{CO}_3/\text{mol Na}_2\text{CO}_3 = 2.053 \text{ g}$

Percent Yield = $\frac{2.053 \text{ g}}{3.255 \text{ g}} \times 100 = 63.10\%$

EQUIPMENT* AND CHEMICALS NEEDED

Safety goggles

Milligram Balance

Spatula

Bunsen burner and tubing

Ring stand and clamp

Large test tube

Test tube holder

Baking soda (Sodium bicarbonate, sodium hydrogen carbonate, NaHCO_3)

* Images of equipment needed in this lab are in the appendix (the equipment may differ slightly or be subject to changes; follow your instructors' directions).

SAFETY PRECAUTIONS

- 1) Wear chemical splash goggles throughout the experiment.
- 2) Gloves are provided if you wish to wear them.
- 3) Please review Material Safety Data Sheets for additional questions.
- 4) Exercise caution when using the Bunsen burner and when handling heated objects. Use extreme caution while using a Bunsen burner and handling a hot ring stand and test tube.
- 5) Do not touch the crucible or any metal that may remain hot. Use heat-resistant gloves if necessary. Discard the waste in the proper waste containers.
- 6) Be sure all glassware is clean and all equipment is returned to its proper place.
- 7) Clean the lab benches and check the laminated sheets to ensure all equipment is in your lab drawer before leaving the lab.
- 8) Wash your hands once you leave the lab.

EXPERIMENTAL PROCEDURE

Percent Yield of Sodium Carbonate, Na_2CO_3 , obtained by decomposition of Baking Soda, NaHCO_3

- 1) Put on safety goggles.
- 2) Set up the Bunsen burner and ring stand as shown in Figure 1.

Set up for thermal decomposition of sodium bicarbonate. Test tube with sodium bicarbonate clamped to ring stand being heated using Bunsen burner.

Figure 1. Set up for thermal decomposition of sodium bicarbonate. A test tube with sodium bicarbonate clamped to a ring stand is heated using a Bunsen burner.

- 3) Weigh a large, clean, empty, dry test tube and record its exact mass.
- 4) Add 1.5-2.0 grams of baking soda into the test tube using a spatula.
- 5) Weigh the test tube with the baking soda and record the combined mass displayed on the milligram balance.

- 6) Attach the test tube with baking soda to the clamp on the ring stand (figure 1) and gently heat the test tube mixture with the Bunsen burner flame.
- 7) You will notice water droplets collecting at the end of the test tube. Heat until no further change is observed, indicating the completion of thermal decomposition.
- 8) Make sure all the moisture is gone before you stop heating. If there is any moisture in the test tube or sample, carefully use the flame to dry it. Any moisture will result in increased mass and lead to incorrect calculations.
- 9) Allow the test tube and its contents to cool to room temperature for 10 -15 minutes.
- 10) Weigh the test tube with the decomposed baking soda, which is the test tube now containing the sodium carbonate, Na_2CO_3 , residue, and record the mass.
- 11) Gently heat this test tube for 10 minutes, cool it to room temperature, and record its mass as the second heating. The masses of sodium carbonate and Na_2CO_3 residue along with the test tube should be almost constant (within 0.05g). If not, a third heating will be needed. Follow the above steps for a third heating.
- 12) Calculate the theoretical and percent yields (refer to example 1) for both trials and calculate the average percent yield in the data table.
- 13) Discard the solid as indicated by your instructor.

PRE-LAB QUESTIONS

Name: _____

1. Define the following:

a) Actual yield:

b) Theoretical yield:

c) Percent yield:

2. What happens if moisture or water remains in the test tube?

3. 2.5g of sodium bicarbonate is equivalent to how many moles of sodium bicarbonate? (Show your work)

4. What theoretical yield and percent yield would be if 2.525 g of sodium bicarbonate is decomposed to 1.487 g sodium carbonate, Na_2CO_3 ? (Show your work*)

DATA AND OBSERVATIONS

Name: _____ Lab Partner(s): _____

Percent Yield of Sodium Carbonate, Na_2CO_3 , obtained by decomposition of Baking Soda, NaHCO_3

1. Mass of empty test tube [m_1] =

2. Mass of test tube + baking soda (before heating) [m_2] =

3. Mass of baking Soda, NaHCO_3 [$m_3 = m_2 - m_1$]* =

4. Mass of test tube and sodium carbonate, Na_2CO_3 (After heating first heating) [m_4] =

5. Mass of test tube and sodium carbonate, Na_2CO_3 (After heating second heating) [m_5] =

Mass of test tube and sodium carbonate, Na_2CO_3 (After heating third heating if needed) [m_6] =

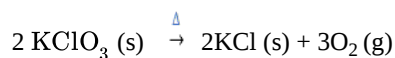
6. Mass of sodium carbonate, Na_2CO_3 [$m = m_5 - m_1$] or [$m = m_6 - m_1$]* =

7. Theoretical yield of Na_2CO_3 (Refer to Example 1), show your work*.

8. Percent yield of Na_2CO_3 (Refer to Example 1), show your work*.

POST-LAB QUESTIONS

1. A student stopped the reaction early without ensuring the decomposition of the sample of sodium bicarbonate, NaHCO_3 , in the test tube. How would this affect the results? Explain.
2. 8.00 g of sodium bicarbonate, NaHCO_3 when decomposed produced 4.257 g of Na_2CO_3 . What is the percent yield of Na_2CO_3 ? (Show your work*)
3. Potassium chlorate, KClO_3 , decomposed according to the following equation.



What mass of potassium chloride would be theoretically produced if 25.253 g of KClO_3 ? (Show your work*)

*Show all your calculations and work where possible for full credit.

Please click [here](#) to access the Pre-Lab, Data Tables, and Post-Lab in Word or PDF format. Complete them and upload according to your instructor's instructions.

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10: Molecular Geometries

11.1: Molecular Geometries – 3D Printing

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10.1: LAB 10 A - MOLECULAR GEOMETRIES – 3D Printing

PURPOSE

The purpose of this experiment is to

- Use the concepts of Lewis structures and VSEPR to predict molecular geometry.
- Create and print 3D models of representative molecules and compare the shapes with predictions.
- Use the 3D models to help predict whether the molecule will be polar or nonpolar.

INTRODUCTION

A molecule's 3-dimensional structure is an important predictor of its physical and chemical properties. One important property of a molecule is its polarity. Most covalent bonds are polar; however, the overall polarity of a molecule will depend on its 3-dimensional geometry. Given a drawing of a molecule, its 3D structure can sometimes be difficult to imagine, making predictions about its polarity problematic. The 3D structure can be evaluated much easier with the aid of a small 3D model, such as one that can be created by a 3D printer.

SAFETY PRECAUTIONS

Follow all your instructor's instructions for [properly using](#) your 3D printer. Parts of the printer can be hot, and the mechanical moving parts can cause injury unless proper care is taken.

EQUIPMENT* AND CHEMICALS NEEDED

- 3D printer
- Computer with the 3D chemical structure drawing and 3D printing software installed (described below)

* Images of equipment needed in this lab can be found in the appendix (the equipment may differ a little or be subject to changes; follow your instructors' directions).

EXPERIMENTAL PROCEDURE

The following programs will be used for the instructions for this lab. These are all available free of charge.

- Jmol[1] to create a 3D computer model of a molecule and export it as an STL file.
- Ultimaker CURA[2] to slice the STL file and create the g-code needed by the 3D printer.

For each of the compounds in the data table, you will

1. create a 3D computer model of a molecule of the compound.
2. export an STL file that can be "sliced" by CURA.
3. import the STL file into CURA, slice it, and generate the g-code.
4. move the g-code to the printer and create a 3D print of the molecule.
5. use the model to help predict the molecule's polarity.

Complete the following Preliminary Exercise to practice the steps involved.

Preliminary Exercise – Printing a 3D model of carbon tetrafluoride, CF₄.

Complete the following steps to build and print your first model.

1. Create a 3D computer model.
 - a. Open Jmol by clicking on the Jmol.jar file. This will open the 3D window of the application.

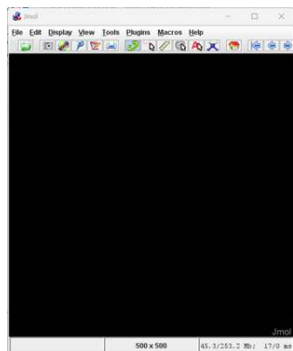


Figure 11.1: A screenshot of the Jmol Application.

- b. Select Get MOL from the File menu. You can enter each of the molecules in the Data Table as a formula (e.g., CF₄) or by name (e.g., tetrafluoromethane). Sometimes the formula will not work. If you have difficulties, ask your instructor for assistance.

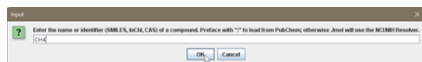
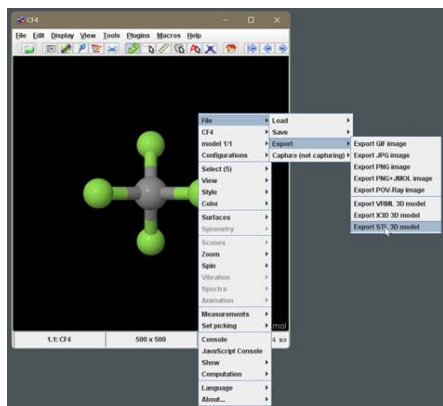


Figure 11.2: A screenshot of an Input dialog box

c. When you press enter or click on OK, a 3D model of the molecule will be displayed in the window.

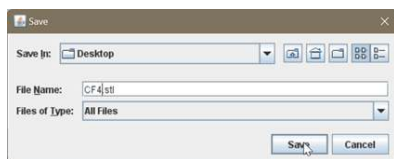
2. Export an STL file.

a. Right click on the Jmol display window and open successive menus to select File ► Export ► Export STL 3D model.



A computer screen shot of a computer Description automatically generated

b. Choose a folder and give your file a name with the .stl extension.



A screenshot of a computer Description automatically generated

3. Slice the STL file in CURA.

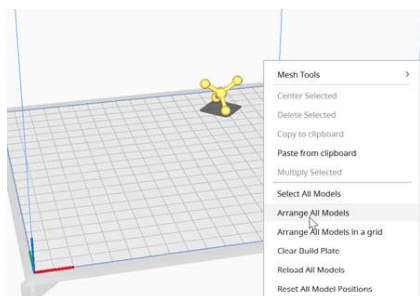
a. Open CURA then click on the open file icon.



A screenshot of a computer Description automatically generated

b. Browse to where you saved the STL file and open it.

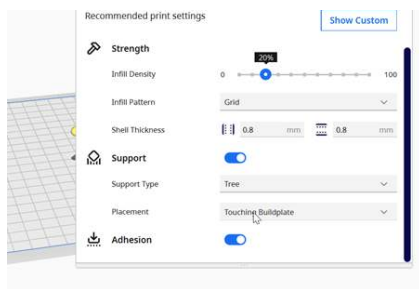
c. The model will be displayed on the printing surface. Right click on the printing surface and select "Arrange all models" to move your model to the center of the surface.



A screenshot of a computer Description automatically generated

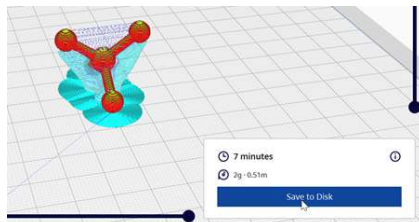
d. You can do various manipulations to the 3D model, such as scaling it and tilting it, but usually, the default will work well. Ask your instructor for help if you are unsure.

e. Make sure that you include [supports](#) for your print. The Tree type supports work well with these models and usually selecting "Touching Buildplate" will be sufficient. If you have trouble with the print, try "Everywhere" for your support placement and ask your instructor for assistance.



A screenshot of a computer Description automatically generated

f. Click on the SLICE button. When the slicing is done, the program will display the amount of time needed to print the model. Click on the “Preview” button to see what the print and supports will look like. Get your instructor’s approval before continuing.



A screenshot of a computer generated model Description automatically generated

g. Export the G-code by clicking on the Save to Disk button.

h. Copy the .gcode file that you just saved to an appropriate removable card as per your instructor.

4. Print the model.

a. Insert the memory card in your printer and use the printer’s control knobs to select your model. The printer will start warming up the nozzle and should begin printing within a few minutes.

b. Once the printing is complete, carefully remove the model and its supports from the printing surface.

c. The support should be carefully removed using needle-nose pliers or other tools provided by your instructor.

5. Predict the polarity of the molecule.

a. Look at the model carefully and rotate the model to answer these questions:

Does the molecule contain polar bonds?

In what direction(s) are the polar bonds?

Are the bonds symmetrically oriented so that their individual polarities cancel each other out? Or are they asymmetrically oriented so that there will be a net dipole in the molecule?

b. Enter the observations and conclusions in the Data Table.

PRE-LAB QUESTIONS

Name: _____

- Describe the similarities and differences between ionic bonds, polar covalent bonds, and pure covalent bonds.
- Explain in your own words how it can be determined if a molecule containing polar bonds is polar or nonpolar.
- Draw the Lewis structure of a molecule of each of the compounds in the Data Table and use Valence Shell Electron-Pair Repulsion (VSEPR) theory to predict the molecule’s geometry. Fill in the appropriate spaces on the table with your answers.

DATA AND OBSERVATIONS

Name: _____ Lab Partner(s): _____

Molecule	CO ₂	
# valence electrons		Lewis Structure
# of electron domains		
Electron domain geometry		
# of lone pairs		3D Sketch /Image of the Model
Molecular geometry		

Polar or nonpolar?

Molecule	H ₂ CO	
# valence electrons		Lewis Structure
# of electron domains		
Electron domain geometry		
# of lone pairs		3D Sketch/ Image of the Model
Molecular geometry		
Polar or nonpolar?		

Molecule	SO ₂	
# valence electrons		Lewis Structure
# of electron domains		
Electron domain geometry		
# of lone pairs		3D Sketch/ Image of the Model
Molecular geometry		
Polar or nonpolar?		

Molecule	CF ₄	
# valence electrons		Lewis Structure
# of electron domains		
Electron domain geometry		
# of lone pairs		3D Sketch/ Image of the Model
Molecular geometry		
Polar or nonpolar?		

Molecule	NH ₃	
# valence electrons		Lewis Structure
# of electron domains		
Electron domain geometry		
# of lone pairs		3D Sketch/ Image of the Model
Molecular geometry		
Polar or nonpolar?		

Molecule	H ₂ O	
# valence electrons		Lewis Structure
# of electron domains		

Electron domain geometry		3D Sketch/ Image of the Model
# of lone pairs		
Molecular geometry		
Polar or nonpolar?		

Molecule	BF ₃	
# valence electrons		Lewis Structure
# of electron domains		
Electron domain geometry		
# of lone pairs		3D Sketch/ Image of the Model
Molecular geometry		
Polar or nonpolar?		

Molecule	PCl ₅	
# valence electrons		Lewis Structure
# of electron domains		
Electron domain geometry		
# of lone pairs		3D Sketch/ Image of the Model
Molecular geometry		
Polar or nonpolar?		

Molecule	IF ₅	
# valence electrons		Lewis Structure
# of electron domains		
Electron domain geometry		
# of lone pairs		3D Sketch/ Image of the Model
Molecular geometry		
Polar or nonpolar?		

POST-LAB QUESTIONS

1. How does the 3D model help you to determine if a molecule is polar or nonpolar?
2. Is the 3D model different than you expected in any way? If so, how?
3. What are the advantages and disadvantages of 3D printing the model vs. looking at the model on a computer screen?

Please click [here](#) to access the Pre-Lab, Data Tables, and Post-Lab in Word or PDF format. Complete them and upload according to your instructor's instructions.

ADDITIONAL INFORMATION REGARDING SOFTWARE

Here are some suggested software packages that can be used to complete this lab. These are all free of charge.

- A program to create and save a stereolithography file, known as an STL file. This is a 3D solid model that can be “sliced” and printed.

- Jmol (requires java): <https://sourceforge.net/projects/jmol/files/Jmol/>. The download consists of a zip file containing all the files needed to run the program. Simply extract the files to a folder of your choice. The program is started by clicking the Jmol.jar file.
- Note: If Java is not available on your computer, an online version can be accessed on the Hack-a-mol website (<https://chemapps.stolaf.edu/jmol/jsmol/hackamol.htm>)
- A slicer program to prepare the STL file for printing and generate the g-code needed by the printer to make the print.
 - Your printer may come with a slicer program, in which case it should be used.
 - One of the best is Ultimaker CURA (<https://ultimaker.com/software/ultimaker-cura/>) The program must be installed on your computer, so some administrative rights are required.
 - There are also several free sites that can generate g-code from an STL file online. An example is Creaity Cloud: <https://www.creaitycloud.com/blog/tutorials/how-to-convert-stl-to-gcode>. Many others can be found by searching the web.

[1] Jmol: an open-source Java viewer for chemical structures in 3D. <http://www.jmol.org/>

[2] <https://ultimaker.com/software/ultimaker-cura/>

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10.2: LAB 10 B - MOLECULAR GEOMETRIES - MODELING KIT

PURPOSE

The purpose of this experiment is to:

- Use the concepts of Lewis structures and VSEPR to predict molecular geometry.
- Create 3D models of representative molecules and compare the shapes with predictions.
- Use the 3D models to help predict whether the molecule will be polar or nonpolar.

INTRODUCTION

A molecule's 3-dimensional structure is one of the key predictors of its physical and chemical properties. One important property is its polarity. Most covalent bonds are polar; however, the overall polarity of a molecule depends on its 3D geometry. When given a drawing of a molecule, visualizing its 3D structure can be challenging, making predictions about its polarity difficult. The 3D structure can be more easily evaluated by building models with a modeling kit or using a small 3D model, such as one created by a 3D printer.

SAFETY PRECAUTIONS

Follow all your instructor's instructions to effectively use your model kit or 3-D printer, and return all the models and equipment as instructed.

EQUIPMENT* NEEDED

- Model kit / 3-D printer (if using a 3-D printer, please see the other procedure)
- A combination of both can also be used. Students can use the model kits to complete the data while their 3-D models are being printed.

* Images of equipment needed in this lab can be found in the appendix (the equipment may differ a little or be subject to changes; follow your instructors' directions).

EXPERIMENTAL PROCEDURE

1. Draw the Lewis structures and use VSEPR theory to predict molecular geometry and whether the molecule will be polar or nonpolar.
2. Create 3D models of the molecules (using a Modeling Kit or 3D printer; if using a 3D printer, please see the other procedure for printing the models).
3. Compare the shapes and use the 3D models to help determine if the molecule will be polar or nonpolar.

Create models of the molecules listed in the data table using the modeling kit provided by your instructor. Include the image in the data table or upload them separately, ensuring they are labeled correctly if uploaded separately.

PRE-LAB QUESTIONS

Name: _____

1. Describe the similarities and differences between ionic bonds, polar covalent bonds, and pure covalent bonds.
2. Explain in your own words how it can be determined if a molecule containing polar bonds is polar or nonpolar.
3. Draw the Lewis structure of a molecule of each of the compounds in the Data Table and use Valence Shell Electron-Pair Repulsion (VSEPR) theory to predict the molecule's geometry. Fill in the appropriate spaces on the table with your answers.

DATA AND OBSERVATIONS

Name: _____ Lab Partner(s): _____

Molecule	CO ₂	Lewis Structure
# valence electrons		
# of electron domains		
Electron domain geometry		

# of lone pairs		3D Sketch/Image of the Model
Molecular geometry		
Polar or nonpolar?		

Molecule	H ₂ CO	
# valence electrons		Lewis Structure
# of electron domains		
Electron domain geometry		
# of lone pairs		3D Sketch/ Image of the Model
Molecular geometry		
Polar or nonpolar?		

Molecule	SO ₂	
# valence electrons		Lewis Structure
# of electron domains		
Electron domain geometry		
# of lone pairs		3D Sketch/ Image of the Model
Molecular geometry		
Polar or nonpolar?		

Molecule	CF ₄	
# valence electrons		Lewis Structure
# of electron domains		
Electron domain geometry		
# of lone pairs		3D Sketch/ Image of the Model
Molecular geometry		
Polar or nonpolar?		

Molecule	NH ₃	
# valence electrons		Lewis Structure

# of electron domains		
Electron domain geometry		
# of lone pairs		3D Sketch/ Image of the Model
Molecular geometry		
Polar or nonpolar?		

Molecule	H ₂ O	
# valence electrons		Lewis Structure
# of electron domains		
Electron domain geometry		
# of lone pairs		3D Sketch/ Image of the Model
Molecular geometry		
Polar or nonpolar?		

Molecule	BF ₃	
# valence electrons		Lewis Structure
# of electron domains		
Electron domain geometry		
# of lone pairs		3D Sketch/ Image of the Model
Molecular geometry		
Polar or nonpolar?		

Molecule	PCl ₅	
# valence electrons		Lewis Structure
# of electron domains		
Electron domain geometry		
# of lone pairs		3D Sketch/ Image of the Model
Molecular geometry		
Polar or nonpolar?		

Molecule	IF ₅	
# valence electrons		Lewis Structure
# of electron domains		
Electron domain geometry		
# of lone pairs		3D Sketch/ Image of the Model
Molecular geometry		
Polar or nonpolar?		

POST-LAB QUESTIONS

1. How does the 3D model help you to determine if a molecule is polar or nonpolar?
2. Is the 3D model different from what you expected in any way? If so, how?
3. What are the advantages and disadvantages of a 3D model vs. looking at the model on a computer screen?

Please click [here](#) to access the Pre-Lab, Data Tables, and Post-Lab in Word or PDF format. Complete them and upload according to your instructor's instructions.

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11: LAB 11 - BOYLE'S LAW [Pressure-Volume Relationship in Gases]

PURPOSE

The purpose of this experiment is to:

- A gas pressure sensor and a gas syringe measure the pressure of an air sample at several different volumes.
- Determine the relationship between pressure and volume of the gas.
- Describe the relationship between gas pressure and volume in a mathematical equation.
- Use the results to predict the pressure at other volumes.

INTRODUCTION

British scientist Robert Boyle, in 1662, found that the pressure of a gas is inversely proportional to its volume, provided the number of moles of gas and the temperature remain constant. Mathematically, Boyle's law can be stated as:

$$P \propto 1/V \text{ Pressure is inversely proportional to the volume}$$

Or

$$PV = k \text{ Pressure multiplied by volume equals some constant } k$$

Where P is the pressure of the gas, V is the volume of the gas, and k is a constant.

Boyle's Law states that the product of pressure and volume is constant when the number of moles and temperature of a gas are held constant.

When comparing the same gas under two different sets of conditions of volume and pressure at constant temperature and number of moles, Boyle's law can also be expressed as:

$$P_1V_1 = P_2V_2$$

P_1 and P_2 are the initial changes in gas pressure at volumes V_1 and V_2 , respectively.

The gas pressure can be expressed in different units, such as Atmospheres (atm), Millimeters of mercury (mm Hg), Torr, and Pascals. The standard unit of pressure is the atmosphere, and the SI unit is the pascal (Pa). A pascal is defined as a pressure of one newton per square meter. This unit is inconveniently small for many purposes, and kilopascal (kPa) is more commonly used.

$$1 \text{ atm} = 760 \text{ mm Hg} = 760 \text{ Torr} = 101325 \text{ Pa} = 101.3 \text{ kPa}.$$

The volume of a gas can also be expressed in various units, such as liters (L), milliliters (mL), cubic meters (m^3), and cubic centimeters (cm^3).

In this experiment, you will measure the pressure of a sample of air at several different volumes. You will also graph pressure versus the reciprocal of volume ($1/V$) to demonstrate that pressure is inversely proportional to the volume of a gas. This indicates that as the volume increases, gas pressure decreases proportionally, and vice versa.



Figure 1: Syringe connected to the Vernier gas pressure sensor.

The primary objective of this experiment is to determine the relationship between the pressure and volume of a confined gas. We will use air confined to a syringe connected to a Gas Pressure Sensor.

When the volume of the syringe is changed by moving the piston, a change occurs in the pressure exerted by the confined gas. This pressure change will be monitored using a Gas Pressure Sensor. It is assumed that the temperature will be constant throughout the

experiment. Pressure and volume data pairs will be collected and analyzed during this experiment. From the data and graph, you should be able to determine the mathematical relationship between the pressure and volume of the confined gas.

EQUIPMENT AND CHEMICALS NEEDED

Lab Quest interface or Vernier Computer interface with the Logger Pro software

Gas Pressure Sensor

20 mL gas syringe

* Images of equipment needed in this lab are in the appendix (the equipment may differ slightly or be subject to changes; follow your instructor's directions).



SAFETY PRECAUTIONS

1. Regular safety precautions, including wearing safety goggles and following proper lab procedures.
2. Be sure all equipment is returned to its proper place.
3. Clean the lab benches and check the laminated sheets to ensure all equipment is in your lab drawer before leaving.
4. Wash your hands once you leave the lab.

This lab experiment is adapted from Vernier and uses Logger Pro software, the LabQuest interface, and the Vernier gas pressure sensor.

EXPERIMENTAL PROCEDURE

1. Prepare the Gas Pressure Sensor and an air sample for data collection.

- a. Obtain a LabQuest interface and turn it on by pressing the power button. The device opens on the LabQuest App screen.
- b. Connect the Gas Pressure Sensor to it and choose **New** from the file menu.
- c. Take the 20.0 mL syringe (not yet connected to the Gas Pressure Sensor and move the syringe's piston until the front edge of the inside black ring (indicated by the arrow in Figure 2) is positioned at the 10.0 mL mark.
- d. Attach the 20.0 mL syringe to the valve of the Gas Pressure Sensor.

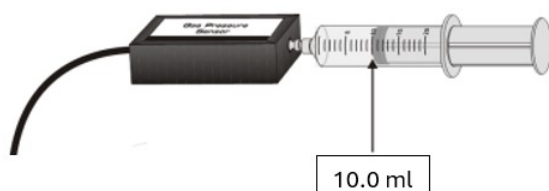


Figure 2: Syringe connected to the vernier gas pressure sensor, with an arrow pointing to 10.0 mL.

2. Set up the data-collection mode.

- a. On the Meter screen, tap Mode. Change the mode to Events with Entry.
 - b. Enter the Name (Volume) and Units (mL). Select OK.
3. To obtain the best data possible, you must correct the volume readings from the syringe. Look at the syringe; its scale reports its internal volume. However, that volume is not the total volume of trapped air in your system, as there is a small space inside the pressure sensor.
- To account for the extra volume in the system, you must add 0.8 mL to your syringe readings. For example, with a 5.0 mL syringe volume, the total volume would be 5.8 mL. You will need this total volume for the analysis.
4. You are now ready to collect pressure and volume data. It is easiest if one person handles the gas syringe and another enters the volumes.

- a. Start data collection by clicking the green arrow.
- b. Move the piston so the front edge of the inside black ring (see Figure 3) is positioned at the 5.0 mL line on the syringe. Hold the piston firmly in this position until the pressure value displayed on the screen stabilizes.
- c. When the pressure reading has stabilized, click Keep. (The person holding the syringe can relax after Keep is clicked.) Type in the total gas volume (in this case, 5.8 mL) in the edit box. Remember, you are adding 0.8 mL to the syringe's total volume. Select OK to store this pressure-volume data pair.

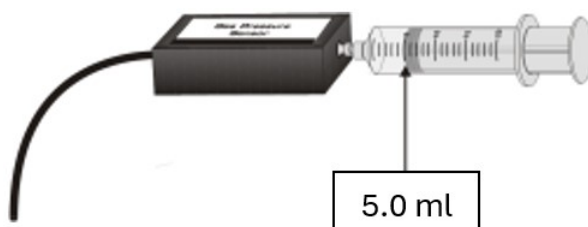


Figure 3: Syringe connected to the Vernier gas pressure sensor, with an arrow pointing to 5.0 mL.

- d. Move the piston to the 7.0 mL line. When the pressure reading has stabilized, click "Keep" and enter the total volume, 7.8 mL.
 - e. Continue this procedure using 10.0, 12.5, 15.0, 17.5, and 20.0 mL syringe volumes. Remember to add 0.8 mL to each syringe volume and enter the total volumes only.
 - f. Click the stop when you have finished collecting data.
5. When data collection is complete, a graph of pressure vs. volume will be displayed. Tap any data point to examine the data pairs displayed in the graph. As you tap each data point, the pressure and volume values are displayed to the right of the graph. Record the pressure and volume data values in your data table.
6. Based on the graph of pressure vs. volume, decide what kind of mathematical relationship exists between these two variables, direct or inverse. To see if you made the right choice:
- a. Choose Curve Fit from the Analyze menu.
 - b. Select Power as the Fit Equation. The curve fit statistics for these two data columns are displayed as an equation.

$$y = Ax^B$$

where **x** is volume, **y** is pressure, **A** is a proportionality constant, and **B** is the exponent of **x** (volume). Note: The relationship between pressure and volume can be determined from the value and sign of the exponent, **B**.

- c. If you have correctly determined the mathematical relationship, the regression line should nearly fit the points on the graph (that is, pass through or near the plotted points).
 - d. Select OK.
7. Once you have confirmed that the graph represents either a direct or inverse relationship, email or print a copy of the graph, with the graph of pressure vs. volume and its best-fit curve displayed, to include in your lab report.
8. A graph of pressure vs. the reciprocal of volume (1/volume) may also be plotted to confirm an inverse relationship between pressure and volume. To do this, it is necessary to create a new column of data, reciprocal of volume, based on your original volume data:
- a. Tap the Table tab to display the data table.
 - b. Choose New Calculated Column from the Table menu.

- c. Enter the Name (1/Volume) and Units (1/mL). Select the equation, A/X. Use Volume as the Column for X, and one as the value for A. Select OK.
9. Follow this procedure to calculate regression statistics and to plot a best-fit regression line on your graph of pressure vs. 1/volume:
 - a. Choose Graph Options from the Graph menu.
 - b. Select Auto scale from 0 and select OK.
 - c. Choose Curve Fit from the Analyze menu.
 - d. Select Linear as the Fit Equation. The linear-regression statistics for these two data columns are displayed in the form: $y = mx + b$ where x is 1/volume, y is pressure, m is a proportionality constant, and b is the y-intercept.
 - e. Select OK. If the relationship between P and V is inverse, the graph of pressure vs. 1/volume should be direct; that is, the curve should be linear and pass through (or near) the origin. Examine your graph to see if this is true for your data. Print or email a copy of the graph to your lab report.
10. Go to graph one or run one and proceed directly to the Processing the Data section with the displayed best-fit curve.

PRE-LAB QUESTIONS

Name: _____

1. State Boyle's Law and state the relationship between Pressure and Volume of a gas at constant temperature and moles of gas.
2. Students, while conducting Boyle's law experiment, measure that a gas syringe at STP (Standard Temperature and Pressure $\equiv$ 1.0 atm; 273 K) has a volume of 20.0 ml if they push the plunger to 10.0 ml, what would be the pressure exerted by the air in the syringe (no change in the amount of air trapped in the syringe and the temperature)(Show your work)
3. Convert the following: (Show your work)
 - a) 0.85 atm = _____ kPa
 - b) 650 mmHg = _____ atm = _____ torr = _____ kpa
 - c) 0.250 L = _____ mL = _____ cm^3

DATA, OBSERVATIONS, AND CALCULATIONS

Name _____ Lab Partner(s) _____

Table 11.2: Relationship between pressure, P, and volume, V

Volume (mL)	Pressure (kPa)	Constant, k (P / V or $P \cdot V$) [*] Show your work

Volume (mL)	Pressure (kPa)	Constant, k (P / V or $P \cdot V$)* Show your work

* One way to determine if a relationship between pressure, P , and volume, V , is inverse or direct is to find a proportionality constant, k , from the data. If this relationship is direct, $k = P/V$. If it is inverse, $k = P \cdot V$.

The report should include two graphs, P vs V and P vs $1/V$, along with the above data, calculations, and data analysis questions below.

DATA ANALYSIS AND POST-LAB QUESTIONS

1. With the best-fit curve still displayed on graph one or run one graph, choose Interpolate from the Analyze menu. A vertical cursor now appears on the graph. The cursor's volume and pressure coordinates are displayed in the floating box. Move the cursor along the regression line until the volume value is 5.0 mL. Note the corresponding pressure value. Move the cursor until the volume value is doubled (10.0 mL). What does your data show happened to the pressure when the volume is doubled? Mention the actual pressure values in your answer.
2. Using the same technique as in Question 1, what does your data show happens to the pressure if the volume is halved from 20.0 mL to 10.0 mL? Mention the actual pressure values in your answer.
3. Using the same technique as in Question 1, what does your data show happens to the pressure if the volume is tripled from 5.0 mL to 15.0 mL? Mention the actual pressure values in your answer.
4. From your answers to the first three questions and the shape of the curve in the plot of pressure vs. volume, do you think the relationship between the pressure and volume of a confined gas is direct or inverse? Explain your answer.
5. Based on your data, what would you expect the pressure to be if the syringe volume were increased to 40.0 mL? Explain or show work to support your answer.
6. Based on your data, what would you expect the pressure to be if the syringe volume were decreased to 2.5 mL? Explain or show work to support your answer.

7. What experimental factors are assumed to be constant in this experiment?
8. One way to determine if a relationship is inverse or direct is to find a proportionality constant, k , from the data. If this relationship is direct, $k = P/V$. If it is inverse, $k = P \cdot V$. Based on your answer to Question 4, choose one of these formulas and calculate k for the seven ordered pairs in your data table (divide or multiply the P and V values). Mention which relationship you are using and show the answers in the third column of the Data and Calculations table.
9. How constant were the values for k that you obtained in Question 8? Good data may show some minor variation, but the values for k should be relatively constant.
10. Write an equation representing Boyle's law using P , V , and k . Write a verbal statement that correctly expresses Boyle's law.

Please click here to access the Pre-Lab, Data Tables, and Post-Lab in Word or PDF format. Complete them and upload according to your instructor's instructions.

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12: LAB 12 - AN INVESTIGATION OF INTERMOLECULAR FORCES

PURPOSE

The purpose of this experiment is to:

- Identify intermolecular forces present in the given liquids.
- Test the solubility of various liquids in water.
- Explore the evaporation of liquids and relate the results to the strengths of intermolecular forces.

INTRODUCTION

Intermolecular forces are attractive forces between atoms and molecules in a substance. The three most important intermolecular forces are London dispersion forces, dipole-dipole, and hydrogen bonding. We first look at the polarity to determine the primary intermolecular force present in a substance. Polar substances will have dipole-dipole or hydrogen bonding as the primary intermolecular force, and non-polar substances will have London dispersion forces.

Intermolecular forces play a pivotal role in determining the physical properties exhibited by substances. For instance, they can explain why certain substances form solutions with others. Understanding these forces also enables us to predict trends in boiling points and evaporation rates of liquids based on their strengths.

In this experiment, you will first identify the most important intermolecular force in various liquids by examining their structure and polarity. You will then be asked to perform a literature search to determine the boiling point of each liquid. Finally, you will explore how intermolecular forces influence solution formation and evaporation.

SAFETY PRECAUTIONS

- 1) Always wear chemical splash goggles while working on this experiment.
- 2) Gloves are provided for you if you wish to wear them.
- 3) Work with all chemicals in a well-ventilated area or under a fume hood.
- 4) At the end of the experiment, dispose of all used chemicals in the appropriate waste container.
- 5) Wash and return all glassware to the appropriate place.
- 6) Thoroughly clean your work area before leaving the lab.
- 7) Wash your hands as soon as you leave the lab.

EQUIPMENT* AND CHEMICALS NEEDED

Table 12.1: Equipment and Chemicals

EQUIPMENT	CHEMICALS	CHEMICALS
Well plate	Deionized water	Ethyl acetate
600 mL beaker	Methanol	Butyl acetate
7 small test tubes	Ethanol	Hexane
Test tube rack	Isopropyl alcohol	Non-halogenated organic waste container
Thermometer	1-butanol	-

* Images of equipment needed in this lab can be found in the appendix (the equipment may differ or be subject to changes; follow your instructor's directions).

EXPERIMENTAL PROCEDURE

Part A: Properties of liquids

For each of the liquids used in this experiment, draw the appropriate Lewis structure, predict whether the substance is polar or non-polar, determine the molar mass, identify the most important intermolecular force present, and do a literature search to find the boiling point (in °C). Record all this information in the data table.

Part B: Solubility

Obtain seven small test tubes and a test tube rack. To each test tube, add 1 mL of deionized water. Then, add 1 mL of the given liquids to the following test tubes:

Test tube 1 = methanol

Test tube 2 = ethanol

Test tube 3 = isopropyl alcohol

Test tube 4 = 1-butanol

Test tube 5 = ethyl acetate

Test tube 6 = butyl acetate

Test tube 7 = hexane

Gently shake each test tube and determine whether each liquid is soluble in water. Record your observations in the data table. Dispose of the contents of each test tube in the non-halogenated organic waste container.

Part C: Evaporation

- 1) Half fill wells in a Well plate with each liquid (deionized water, methanol, ethanol, isopropyl alcohol, 1-butanol, ethyl acetate, butyl acetate, and hexane).
- 2) Determine the initial temperature of the deionized water using a Celsius thermometer or the temperature probe with LabQuest. Be sure to *hold the thermometer in the liquid* until the reading stabilizes.
- 3) Once the initial temperature has been recorded, remove the thermometer from the water and *hold it in the air*. Once the temperature stabilizes, *record the final temperature*.
- 4) Repeat this procedure for each liquid.
- 5) Once you have completed this procedure, pour the contents of the well plate into a 600 mL beaker and then transfer the beaker's contents into the non-halogenated organic waste container.

PRE-LAB QUESTIONS

Name _____

1. Define the following terms:

Part A: Intermolecular forces

Part B: London dispersion forces

Part C: Dipole-dipole

Part D: Hydrogen bonding

2. Match and write the correct intermolecular force (London dispersion force, dipole-dipole, or hydrogen bonding) to each description.

Part A: Intermolecular force that is present when hydrogen is directly bonded to oxygen, nitrogen, or fluorine:

Part B: The strongest of the intermolecular forces:

Part C: The weakest of the intermolecular forces:

Part D: The most important intermolecular force present in HCl:

Part E: Intermolecular force based on a temporary attractive force:

3. What are three factors that affect the strength of intermolecular forces?

4. Circle the best choice (indicated in the parenthesis) for each question:

Part A: Substances with stronger intermolecular forces tend to have (**higher/lower**) boiling points and evaporate (**more readily/less readily**) than substances with weaker intermolecular forces.

Part B: Substances with similar intermolecular forces tend to be (**miscible/immiscible**) with each other.

DATA AND OBSERVATIONS

Name _____ Lab Partner(s) _____

Part A: Properties of Liquids

Table 12.2: PROPERTIES OF LIQUIDS

*Substance	Molar Mass	Boiling Point	Lewis Structure	Polar or Nonpolar	Pri
Water [H ₂ O] HOH					
Methanol [CH ₄ O] CH ₃ OH					
Ethanol [C ₂ H ₅ OH] CH ₃ CH ₂ OH					
Isopropyl Alcohol [C ₃ H ₈ O] CH ₃ CHOHCH ₃					
1-butanol [C ₄ H ₉ -OH] (CH ₃ (CH ₂) ₂ CH ₂ -OH)					
Ethyl Acetate [C ₄ H ₈ O ₂] CH ₃ COOCH ₂ CH ₃					
Butyl acetate [C ₆ H ₁₂ O ₂] CH ₃ COO(CH ₂) ₃ CH ₃					
Hexane [C ₆ H ₁₄] CH ₃ (CH ₂) ₄ CH ₃					

*Name, [Chemical Formula], and Condensed Formula (Showing Connectivity).

Part B: Solubility

Table 12.3: SOLUBILITY IN WATER

Liquid	Soluble or Insoluble in Water?
1-butanol	
Butyl acetate	
Ethanol	
Ethyl acetate	
Hexane	
Isopropyl alcohol	
Methanol	

Part C: Evaporation

Table 12.4: CHANGE IN TEMPERATURE DURING EVAPORATION

Liquid	Initial Temperature	Final Thermometer Reading	Change in Temperature
Water			
Methanol			
Ethanol			
Isopropyl alcohol			
1-butanol			
Ethyl acetate			
Butyl acetate			
Hexane			

POST-LAB QUESTIONS

1. Refer to the results from part A. What is the relationship (if any) between the molar masses and boiling points of the liquids?
2. Refer to the results from part A. Do you notice a trend between the boiling points and primary intermolecular forces in the liquids?
3. Refer to the results from part B. What type of substances were soluble in water? How can you use the concept of intermolecular forces to explain this?
4. Refer to the results from part B. Explain why this was the case for each substance insoluble in water.
5. Write a summary of the results from part C.
6. Is there a connection between the observed evaporation results of the liquids in part C and the boiling points recorded in part A? Explain.
7. Based on the results from part C, is evaporation an endothermic or exothermic process? Explain.
8. Based on the results of this experiment, what can you conclude about the relationship between the strengths of intermolecular forces and the boiling points/evaporation of a liquid?

Please click here to access the Pre-Lab, Data Tables, and Post-Lab in Word or PDF format. Complete them and upload according to your instructor's instructions.

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